

# **Information Retrieval Notes**

Course Notes

Academic Year 2026/2027

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# Chapter 0

## Introduction and Architecture of Information Retrieval

### 0.1 Definition and Motivations

**Information Retrieval** (IR) is the scientific and engineering discipline concerned with representing, storing, organizing, and accessing large collections of predominantly unstructured or semi-structured data to satisfy an **information need** expressed by a user [1, 2].

**Definition 0.1.1** (Information Retrieval (Manning et al.)). Information Retrieval is finding material of an unstructured nature (usually text) that satisfies an information need from within large collections stored on computers.

IR is not limited to general-purpose Web search engines (Google, Bing): it underpins e-commerce and marketplace search, enterprise/desktop search, **RAG** (*Retrieval-Augmented Generation*) systems feeding Large Language Models with contextually relevant external documents, and recommender engines.

#### 0.1.1 From Web Search to Agentic Search

The rise of conversational LLMs has not made Information Retrieval obsolete but has broadened its impact: modern LLMs continuously rely on core IR competencies (semantic query understanding, matching and ranking, contextual retrieval) to overcome their *knowledge cutoff*. In the **RAG** paradigm, the system intercepts the user’s request, queries an IR engine, injects the retrieved documents into the LLM’s context, and generates a response supported by verifiable citations.

**Definition 0.1.2** (Agentic Search). **Agentic Search** is a dynamic retrieval process in which an intelligent agent autonomously decides which subsequent searches to conduct, based on partial evidence gathered: it inspects sources, identifies knowledge gaps (*information gaps*), formulates refined follow-up queries, and iterates until sufficient evidence has converged.

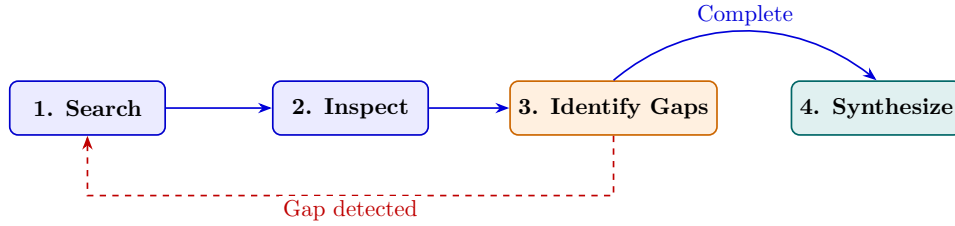


Figure 1: Adaptive feedback loop characterizing Agentic Search.

### 0.1.2 Information Retrieval vs. Data Retrieval

| Dimension              | Data Retrieval (SQL Databases)                   | Information Retrieval                             |
|------------------------|--------------------------------------------------|---------------------------------------------------|
| <b>Data structure</b>  | Rigidly schematized (tables, types, constraints) | Unstructured or semi-structured (free text, HTML) |
| <b>Query language</b>  | Formal and deterministic (SQL)                   | Natural or keyword-based, often ambiguous         |
| <b>Matching</b>        | Exact Boolean                                    | Partial and probabilistic (degrees of relevance)  |
| <b>Ordering</b>        | None unless explicit (ORDER BY)                  | Fundamental, driven by relevance estimation       |
| <b>Error tolerance</b> | None                                             | High (typos, synonymy, polysemy)                  |

Table 1: Comparison between traditional Data Retrieval and Information Retrieval.

#### Relevance: The Cornerstone of IR

While in a database a record either satisfies or violates a Boolean predicate, in Information Retrieval the core challenge is estimating the degree of **relevance** with respect to the user’s information need: subjective, context-dependent, and evolving over time.

## 0.2 The Two Macro-Objectives: Effectiveness and Efficiency

A large-scale retrieval system pursues two objectives simultaneously, often in tension:

1. **Effectiveness:** how accurately the ranking satisfies the user’s information need.
2. **Efficiency:** computational cost, throughput, and response latency, typically on the order of 10–50 milliseconds per query over collections of billions of documents.

The efficiency of a modern search engine depends on its interaction with the underlying hardware architecture: pipelining and superscalar processors, memory hierarchy and access locality, SIMD vectorization, and parallel/distributed computing across clusters (*sharding*).

## 0.3 Architecture of an Information Retrieval System

An IR system enforces a strict decoupling between the **Offline Pipeline** (indexing and ranking model training) and the **Online Pipeline** (low-latency query serving), as depicted in Figure 2.

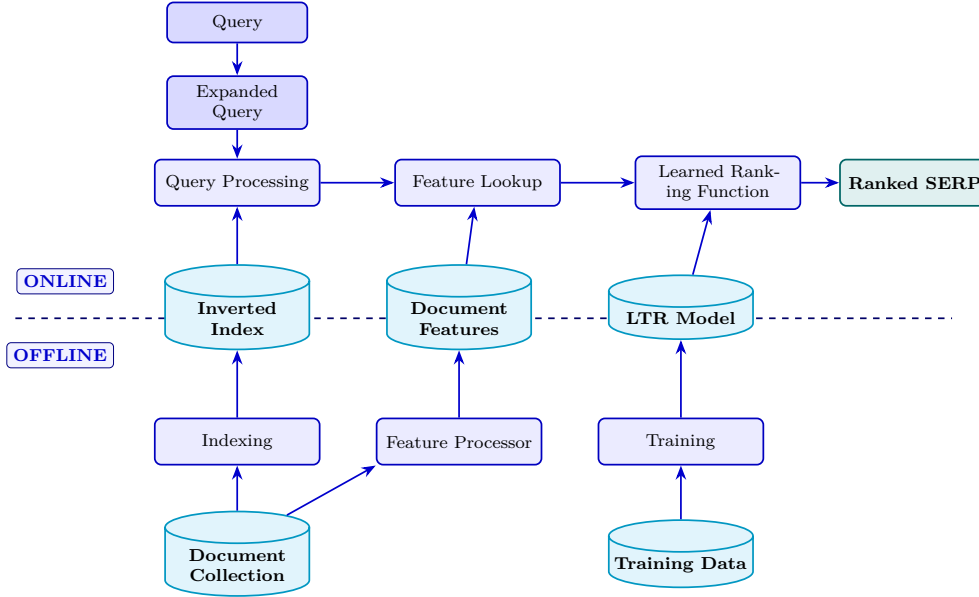


Figure 2: Grid architecture of a modern search engine (Online vs. Offline pipeline).

### 0.3.1 Text Processing Pipeline

Before indexing, every text stream goes through: **tokenization** (segmentation into lexical tokens), **normalization/case folding** (lowercasing, diacritics), **stop word filtering** (removal of high-frequency, low-discriminative words), and **stemming/lemmatization** (reducing an inflected form to its root or canonical lemma, e.g. "searching" → "search").

## 0.4 Language Properties and Retrieval Models

- **Zipf's Law:** the frequency  $f(r)$  of a term is inversely proportional to its rank  $r$  in the frequency table,  $f(r) \propto 1/r$ : a few very frequent words account for most occurrences, while most terms occur only rarely (*long tail*).
- **Heaps' Law:** vocabulary size grows as a power law of the total number of words  $N$  in the collection,  $|V| = k \cdot N^\beta$  (with  $0.4 \leq \beta \leq 0.6$ ): the vocabulary never fully saturates.

The main retrieval models are the **Boolean Model** (exact match via logical operators, no relevance score), the **Vector Space Model** (queries and documents as TF-IDF weighted vectors, relevance via cosine similarity), and the **Probabilistic BM25 Model**, the de facto standard for lexical retrieval:

$$\text{score}_{\text{BM25}}(q, d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{tf(t, d) \cdot (k_1 + 1)}{tf(t, d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)} \quad (1)$$

where  $\text{IDF}(t) = \ln\left(\frac{N - df_t + 0.5}{df_t + 0.5} + 1\right)$  weighs term specificity,  $k_1$  (typically 1.2–2.0) controls term-frequency saturation, and  $b \in [0, 1]$  (typically  $\approx 0.75$ ) normalizes document length  $|d|$  against the average length  $\text{avgdl}$ .

## 0.5 The Inverted Index

The **Inverted Index** reverses the logical orientation of document storage, realizing a mapping  $t \mapsto \mathcal{P}_t$ , where  $\mathcal{P}_t$  is the sorted list of documents containing term  $t$  [1].

**Definition 0.5.1** (Inverted Index). An Inverted Index consists of a **dictionary** (the sorted set of unique terms, with document frequency  $df_t$  and a pointer to the posting list) and, for each term, a **posting list**: an array sorted by increasing DocID, whose entries may include metadata such as local term frequency ( $tf$ ) and positional offsets.

**Example 0.5.1** (Reference Toy Collection). Consider a small collection of three documents about indexing and text search. Figure 3 shows the Inverted Index built for a few key terms: **a** appears in documents 1, 2, 3; **an** only in document 1; **index** and **inverted** in documents 1, 2; **query** only in document 3.

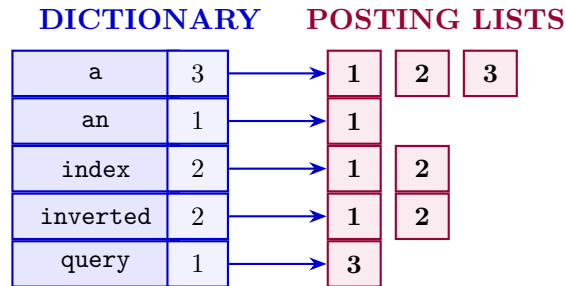


Figure 3: Structure of the Inverted Index: dictionary and posting lists.

Boolean constructs (AND, OR, NOT) and phrase search map directly to the advanced search interface of commercial search engines (Table 2).

| IR Boolean Operator | Google Advanced Search Field     | Syntax                               |
|---------------------|----------------------------------|--------------------------------------|
| Conjunction (AND)   | <i>all these words</i>           | <code>tri-colour rat terrier</code>  |
| Phrase search       | <i>this exact word or phrase</i> | <code>"rat terrier"</code>           |
| Disjunction (OR)    | <i>any of these words</i>        | <code>miniature OR standard</code>   |
| Negation (NOT)      | <i>none of these words</i>       | <code>-rodent -"Jack Russell"</code> |
| Numeric range       | <i>numbers ranging from...to</i> | <code>10..35 kg</code>               |

Table 2: Mapping between IR Boolean operators and Google Advanced Search.

## 0.6 Index Compression Techniques

Inverted Index compression dramatically reduces memory footprint (both in RAM and on disk), converting an I/O bottleneck into a CPU decompression workload, which is generally much faster.

### 0.6.1 Binary Interpolative Coding

**Binary Interpolative Coding** (Moffat and Stuiver [3]) compresses strictly increasing sequences of integers (the DocIDs of a posting list) by recursively partitioning them: given a sublist of  $n$  integers known to lie within  $[l, h]$ , the middle element  $x_m$  (with  $m = \lfloor (n+1)/2 \rfloor$ ) is encoded using  $\lceil \log_2 R \rceil$  bits, where  $R = h - l - n + 1$  is the number of admissible values, and the recursion proceeds separately on the two halves. When DocIDs are densely clustered,  $R$  tends toward 1 and many elements are encoded at near-zero cost.

### 0.6.2 Dictionary-Based Compression: LZ77

For general text, the **Lempel-Ziv '77** (LZ77) algorithm compresses via a sliding window, replacing lexical repetitions with triples composed of (offset, length, next\_char) referring to earlier occurrences in the window. This family of algorithms underlies widely adopted tools such as `gzip`, `7-Zip`, and `LZ4`.

## 0.7 Learning-to-Rank (LTR) and the Non-Differentiability Challenge

Candidates retrieved by the Inverted Index are passed to a machine learning re-ranking stage known as **Learning-to-Rank** (LTR) [4]. The training dataset associates each query with a list of documents carrying graded relevance judgments  $y \in \{0, \dots, 4\}$  (from irrelevant to perfectly relevant); a model computes a score  $s_i = f_\theta(\vec{x}_{q,d_i})$  for each query-document pair, and the final list is obtained by sorting in decreasing order of score.

### The Cornerstone of LTR

**The goal is to learn the ranking, not the label:** the objective is not to accurately estimate the single value  $y$ , but the optimal relative ordering, maximizing cumulative metrics such as **NDCG** or **MAP**.

Because the sorting operation is discrete and step-wise, the gradient of ranking metrics with respect to model parameters is zero almost everywhere: gradient descent cannot be applied directly. The literature has developed three families of solutions:

1. **Pointwise:** treats each document as an independent instance (regression or classification on  $y$ ), ignoring the inherently comparative nature of ranking.
2. **Pairwise** (e.g. RankNet): compares pairs of documents, modeling the probability that  $d_i$  precedes  $d_j$  with a differentiable sigmoid.
3. **Listwise** (e.g. LambdaMART): optimizes the entire list via surrogate gradients  $\lambda_{ij}$ , weighted by the actual NDCG change resulting from swapping  $d_i$  and  $d_j$ ; combined with gradient-boosted decision trees, it was for over a decade the standard in commercial search engines.

## 0.8 The Shift Toward Neural Information Retrieval

Since 2018, Transformer-based models (BERT) have ushered in the era of Neural Information Retrieval [5, 6], pre-trained via **Masked Language Modeling**. Neural retrieval approaches fall into two families:

- **Interaction-based (Cross-Encoders):** query and document are concatenated and jointly processed by a Transformer, capturing deep

interactions at high computational cost ( $\mathcal{O}(|\mathcal{D}| \times \text{Transformer})$ ); used only as second-stage re-rankers over a small candidate set.

- **Representation-based (Bi-Encoders):** query and document are independently projected into a vector space by two distinct encoders; document vectors can be pre-computed offline, reducing online matching to a dot product or an approximate nearest-neighbor search.

Among representation-based models, three families of embeddings are distinguished: **dense single-vector** (a single dense vector per document/query, queried with k-ANN graph indexes such as HNSW), **dense multi-vector** (e.g. ColBERT [5], one vector per token with *Late Interaction*/MaxSim aggregation,  $S(q, d) = \sum_{t_q \in q} \max_{t_d \in d} (\vec{E}_{t_q} \cdot \vec{E}_{t_d}^T)$ ), and **sparse single-vector** (e.g. SPLADE [6], interpretable by design and more space/time efficient than multi-vector).

#### *k*-Approximate Nearest Neighbor Search

For collections of millions of documents represented as high-dimensional dense vectors, exact retrieval of the  $k$  most similar to the query is computationally prohibitive: index structures for **approximate nearest-neighbor search** ( $k$ -ANN) are used instead, trading a minimal loss of accuracy for orders-of-magnitude speedups. **TusKANNy** (<https://tuskanny.com/>), developed by ISTI-CNR and the University of Pisa, is a high-performance open-source example written in Rust.

# Chapter 1

## Evaluation of Information Retrieval Systems

### 1.1 Search Engine Architecture

Before diving into evaluation methodology, it is instructive to understand the high-level block diagram of a modern search engine, since each architectural component introduces measurable quality dimensions.

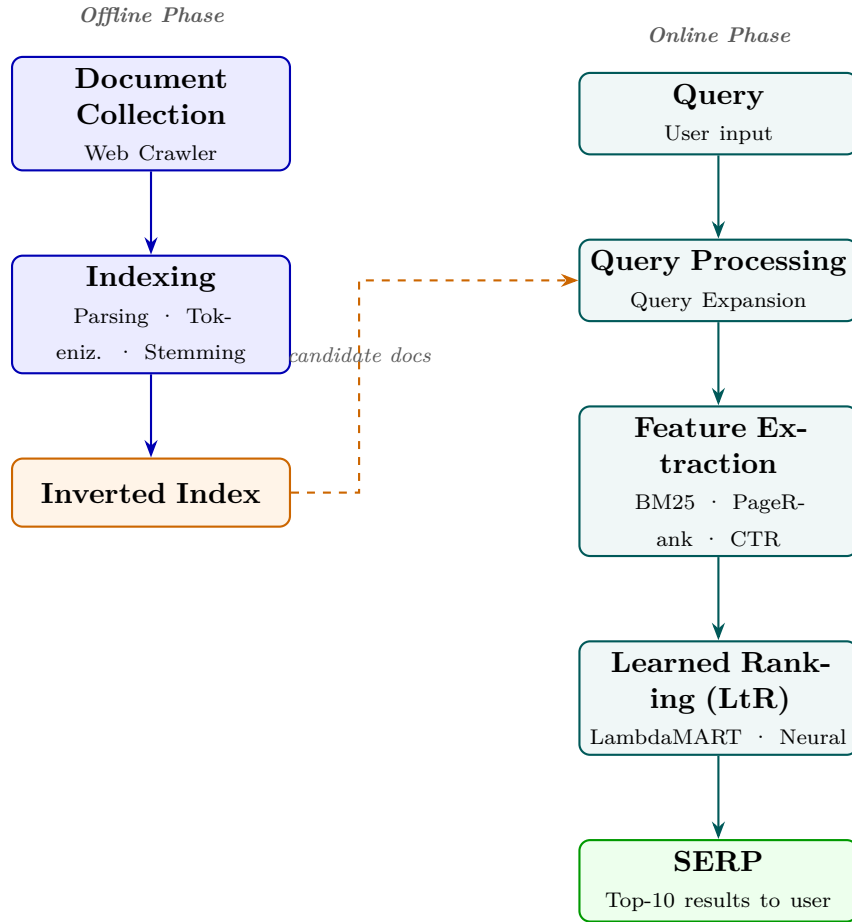


Figure 1.1: High-level block diagram of a modern search engine: offline indexing pipeline (left) and online query processing and ranking pipeline (right).

The main components are:

- **Web Crawler / Document Collection:** An automated agent that continuously discovers and downloads documents from the Web (or a closed corpus), feeding the indexing pipeline.
- **Parsing & Tokenization:** Documents are linguistically analyzed: HTML is stripped, text is tokenized into terms, stop-words may be removed, and stems or lemmas are computed.
- **Inverted Index:** The core data structure mapping each term to the list of documents containing it (posting list), together with positional and frequency information.

- **Query Processing & Query Expansion:** An incoming user query is tokenized with the same pipeline, possibly enriched with synonyms, related terms, or session context to improve recall.
- **Feature Extraction:** For each candidate document, a rich feature vector is assembled (BM25 score, PageRank, freshness, anchor text signals, click history, ...).
- **Learned Ranking Function (Learning to Rank):** A machine-learned model (e.g., LambdaMART, neural ranker) combines the features to produce a final relevance score used to sort the SERP.
- **SERP (Search Engine Results Page):** The final ordered list of results displayed to the user, possibly augmented with featured snippets, knowledge panels, and ads.

## 1.2 Evaluating Search Engine Quality

How do we evaluate such a complex, multi-component system? The answer lies in separating two complementary dimensions—**efficiency** and **effectiveness**—each requiring distinct measurement methodologies [1, 2].

- **Efficiency** concerns the *speed and resource cost* of processing: how fast can we index documents, and how quickly can we answer queries at scale?
- **Effectiveness** concerns the *quality of the answers*: are the returned documents genuinely useful to the user?

Rigorous performance evaluation is essential for designing, fine-tuning, and deploying Information Retrieval systems [1, 2]. Search engine quality is thus evaluated across these two complementary dimensions:

### 1. Efficiency Metrics:

- *Indexing throughput:* Number of documents processed and incorporated per hour.
- *Incremental indexing:* Ingesting new document streams (e.g., +10 000 products daily) without downtime.
- *Search latency and throughput:* Query processing response time (demanding 95th/99th percentile latencies within 10–50 ms) and CPU/memory utilization under workloads of thousands of queries per second over multi-million or multi-billion document collections.

## 2. Effectiveness Metrics and Auxiliary Services:

- Quality of related product recommendations, query spell-checking accuracy, and snippet generation readability.
- **Intrinsic ranking effectiveness:** Measures the engine’s core capability to satisfy users by ranking relevant documents at the top of the Search Engine Results Page (SERP).

## 1.3 Measuring User Happiness

In commercial Web search engines, the overarching objective is maximizing **User Happiness**. Because happiness is an internal cognitive state that cannot be directly queried, production environments rely on observable online behavioral proxies:

- **Clicks and Click-Through Rate (CTR):** The volume and proportion of clicks received by returned results.
  - *Caveat:* Misleading titles and clickbait summaries artificially inflate clicks without truly satisfying the user.
  - *Zero-Click Searches:* When is “no clicks” good news? For factual queries (e.g., weather forecasts, flight schedules, currency conversions, or dictionary definitions), answers rendered directly via *Featured Snippets* or *Knowledge Panels* satisfy the user immediately without requiring any downstream click.
- **Purchases and Economic Conversions:** In e-commerce, the percentage of search sessions resulting in transactions and total monetary spend serve as key indicators.
- **Retention and Repeat Visits:** The fraction of users returning to the search engine within a week or month.
- **Dwell Time:** The duration an attendee spends exploring a clicked page before navigating back to the SERP (*bouncing back*). Extended dwell time indicates genuine engagement, whereas near-instant returns signal dissatisfaction.

## Evaluation in the Era of Generative AI and LLMs

The rise of generative conversational search engines and Large Language Models (LLMs) fundamentally alters traditional metrics such as CTR and Dwell Time: users consume direct synthesized summaries rather than navigating outgoing links. Consequently, factual grounding, hallucination minimization, and source attribution fidelity supplant traditional link click telemetry.

Several critical methodological debates have emerged regarding the evaluation of these systems:

- **Number of conversational turns as a failure proxy:** In a generative dialogue system, an excessive number of turns required to satisfy an information need often signals that the system failed to comprehend the user’s initial intent, forcing repeated clarifications or reformulation loops. Traditional Dwell Time thus yields to conversational length and session efficiency metrics.
- **Prompting variance and reproducibility:** Unlike keyword queries in classical IR, slight changes in prompt wording to a generative LLM can induce substantial variance in response quality. This intrinsic sensitivity severely complicates standardized, reproducible benchmarking.
- **Ontological difference between retrieval and synthesis:** Traditional IR systems *retrieve* and rank preexisting documents from a known collection; generative LLMs *synthesize* original probabilistic text. In retrieval, relevance is evaluated against a fixed corpus and gold-standard ground truth; in synthesis, verifying factual validity and correctness requires entirely new semantic evaluation frameworks.
- **Retrieval-Augmented Generation (RAG):** This paradigm addresses hallucination risks by conditioning generative synthesis on documents fetched via classical IR. Consequently, evaluation requires decoupled yet unified assessment: measuring both retrieval quality and synthesis fidelity.

## 1.4 The Cranfield Paradigm and Offline Test Collections

While online signals offer valuable telemetry, they are noisy, subject to position bias, and unsuitable for comparing algorithmic improvements offline prior to production deployment.

To situate the Cranfield approach historically: Cyril Cleverdon conducted his seminal experiments in the **1960s**, working with *physical* library systems—card catalogues and aeronautical engineering records at the Cranfield Institute of Technology—long before digital text retrieval was computationally feasible. His insight was to construct a controlled experimental collection with exhaustive relevance judgments, providing the first rigorous laboratory for IR comparison. This setting, humble by today’s standards, established a scientific template still followed six decades later.

To establish reproducible, controlled evaluation, IR research adheres to the **Cranfield Paradigm**, pioneered by Cyril Cleverdon in the 1960s [7].

**Definition 1.4.1** (Test Collection). A standard Information Retrieval test collection is a formal triplet  $\mathcal{T} = (\mathcal{D}, \mathcal{Q}, \mathcal{J})$  consisting of:

1. A benchmark document collection  $\mathcal{D} = \{d_1, d_2, \dots, d_N\}$ ;
2. A benchmark suite of queries  $\mathcal{Q} = \{q_1, q_2, \dots, q_M\}$ ;
3. A set of relevance judgments (*qrels*)  $\mathcal{J} = \{(q, d) \mapsto r_{q,d}\}$ , assigning a ground-truth relevance assessment (binary or graded) to each query-document pair.

### 1.4.1 Combinatorial Scale Bottlenecks

Exhaustively assessing all query-document pairs is physically infeasible for realistic corpus sizes. Consider a representative benchmark with:

$$|\mathcal{D}| = 5 \times 10^6 \text{ documents, } |\mathcal{Q}| = 50\,000 \text{ queries}$$

The full Cartesian product comprises:

$$|\mathcal{D}| \times |\mathcal{Q}| = 5 \times 10^6 \times 50\,000 = 2.5 \times 10^{11} \text{ pairs } (0.25 \text{ trillion judgments})$$

Assuming an experienced human assessor evaluates one document in 2.5 seconds:

$$\begin{aligned} \text{Total Time} &= 2.5 \times 10^{11} \times 2.5 \text{ s} = 6.25 \times 10^{11} \text{ seconds} \\ &\approx 173\,611\,111 \text{ person-hours} \end{aligned}$$

This would require tens of thousands of human working years! Consequently, benchmark datasets must restrict assessment to a sampled subset of candidate documents.

### 1.4.2 Origins of Test Queries

A natural question is: *where do the queries in a benchmark test suite come from?* The answer differs significantly depending on the evaluation context:

- **Web search engines — Query Logs:** Large-scale industrial benchmarks (such as those used in Microsoft TREC tracks or the AOL query log dataset) sample queries directly from *real user query logs*. This ensures that the test suite faithfully reflects the actual distribution of user information needs, including rare and long-tail queries.
- **Classic laboratory scenarios — Expert-drafted lists:** In pre-Web IR benchmarks (Cranfield, TREC ad hoc tracks), domain experts *manually composed* a set of information need statements (called *topics*). Each topic is a verbose narrative describing the underlying information need, which is then abbreviated into a short keyword query for system input.
- **What is *not* done — Random queries from documents:** One might naively suggest generating queries by sampling random terms from documents in the collection. This approach is **explicitly avoided** in IR evaluation: randomly sampled terms do not resemble real user intents, would introduce severe artificial biases, and would not generalize to the retrieval scenarios that actually matter to end users.

### 1.4.3 Crowdsourcing: Opportunities and Pitfalls

One cost-effective alternative delegates judging to online micro-task platforms (e.g., Amazon Mechanical Turk). However, empirical literature indicates that crowdsourced annotations suffer from:

- Substantially higher variance compared to domain experts;
- Inconsistent quality and exposure to adversarial spamming;
- A requirement for robust filtering mechanisms (majority voting, worker reputation scoring, Cohen’s  $\kappa$  inter-rater reliability thresholds).

A more recent and increasingly adopted alternative is **LLM-as-a-Judge**:

- **LLM-as-a-Judge:** State-of-the-art language models (such as GPT-4-class models) are prompted with structured evaluation rubrics to assign relevance grades to query-document pairs. This approach offers substantial *economic scalability*—millions of judgments can be obtained at a fraction of the cost of human crowdworkers. However, it introduces well-documented *intrinsic biases*: **length bias** (longer, more verbose responses tend to receive higher scores regardless of content quality), **self-preference bias** (a model tends to rate its own outputs more favorably), and **position bias** (the order in which candidates are presented influences the judgment). Active research is ongoing to quantify and mitigate these distortions.

#### 1.4.4 The TREC Infrastructure

Organized by the National Institute of Standards and Technology (NIST), the **Text REtrieval Conference (TREC)** provides the foundational infrastructure for large-scale IR experimentation. TREC is organized into specialized thematic *tracks* reflecting distinct information access tasks (Web search, biomedical IR, conversational search, deep learning retrieval). The output produced by a participant system executing a retrieval task on a test collection is termed a **Run**.

Table 1.1 documents the dimensional evolution of classical benchmark collections across Information Retrieval history [1], emphasizing the massive scale increase that necessitated the creation of TREC.

| Collection            | Documents ( $N$ ) | Queries    | Size (MB)           | Terms/Doc       | Assessments (Q-D)   |
|-----------------------|-------------------|------------|---------------------|-----------------|---------------------|
| ADI                   | 82                | 35         | —                   | —               | —                   |
| AIT                   | 2 109             | 14         | 2                   | 400.0           | > 10 000            |
| CACM                  | 3 204             | 64         | 2                   | 24.5            | —                   |
| CISI                  | 1 460             | 112        | 2                   | 46.5            | —                   |
| <b>Cranfield</b>      | 1 400             | 225        | 2                   | 53.1            | Exhaustive          |
| LISA                  | 5 872             | 35         | 3                   | —               | —                   |
| Medline               | 1 033             | 30         | 1                   | —               | —                   |
| NPL                   | 11 429            | 93         | 3                   | —               | —                   |
| OSHMED                | 348 566           | 106        | 400                 | 250.0           | 16 140              |
| Reuters               | 21 578            | 672        | 28                  | 131.0           | —                   |
| <b>TREC (Typical)</b> | <b>740 000</b>    | <b>200</b> | <b>2 000 (2 GB)</b> | <b>89–3 543</b> | <b>&gt; 100 000</b> |

Table 1.1: Comparative overview of classical historical evaluation test corpora.

As evident from the empirical data, early corpora from the 1960s and 1970s (such as Cranfield and Medline) comprised only a few thousand records and 1–2 MB, allowing exhaustive manual judging. With the advent of realistic Web-scale corpora containing hundreds of thousands of documents

and gigabytes of text like TREC, the combinatorial evaluation space exploded, mandating selective sampling strategies.

### 1.4.5 The Pooling Technique

To select which documents human assessors should judge without evaluating the entire corpus, TREC standardizes the **Pooling** methodology (Figure 1.2):

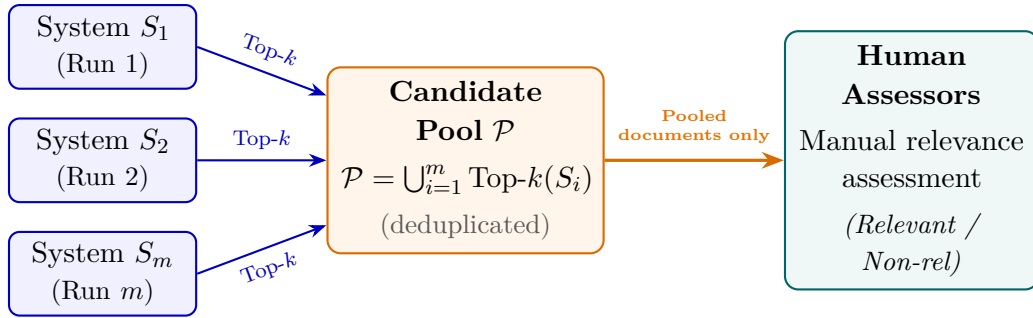


Figure 1.2: The Pooling technique for selecting candidate documents for human judgment.

1. Collect ranked result lists from  $m$  diverse retrieval systems for a given topic.
2. Extract the top- $k$  ranked documents from each run (typically  $k = 50$  or  $k = 100$ ).
3. Take the **union** of these candidate sets, deduplicating documents to build pool  $\mathcal{P}$ .
4. Human judges assess relevance strictly for documents inside  $\mathcal{P}$ .

#### The Unjudged Document Assumption

All documents in the corpus not present in the pool (*unpooled*) are formally assumed to be **non-relevant** ( $r = 0$ ). While this assumption may theoretically penalize innovative systems surfacing relevant documents overlooked by historical pool contributors, NIST experiments confirm that sufficiently diverse pools maintain stable, fair, and reliable system rankings.

## 1.5 Information Need vs. Query

A crucial principle in evaluation separates the underlying need from its textual expression:

- **Information Need:** The user's actual cognitive desire. Example: "*My swimming pool bottom is becoming black and needs to be cleaned*".
- **Query:** The keyword shorthand submitted to the system. Example: "pool cleaner".

**Property 1.5.1** (Evaluation Principle). *Document relevance must always be assessed with respect to the user's underlying **Information Need**, never purely against the superficial keywords of the query string.*

## 1.6 Unranked Retrieval Measures

In unranked evaluation contexts (e.g., Boolean retrieval), the engine returns an unordered set of documents. Performance is derived from the standard contingency table (Table 1.2):

|                        | Document Relevant            | Document Non-Relevant        |
|------------------------|------------------------------|------------------------------|
| Document Retrieved     | True Positive ( <i>TP</i> )  | False Positive ( <i>FP</i> ) |
| Document Not Retrieved | False Negative ( <i>FN</i> ) | True Negative ( <i>TN</i> )  |

Table 1.2: Contingency matrix for binary query-document classification.

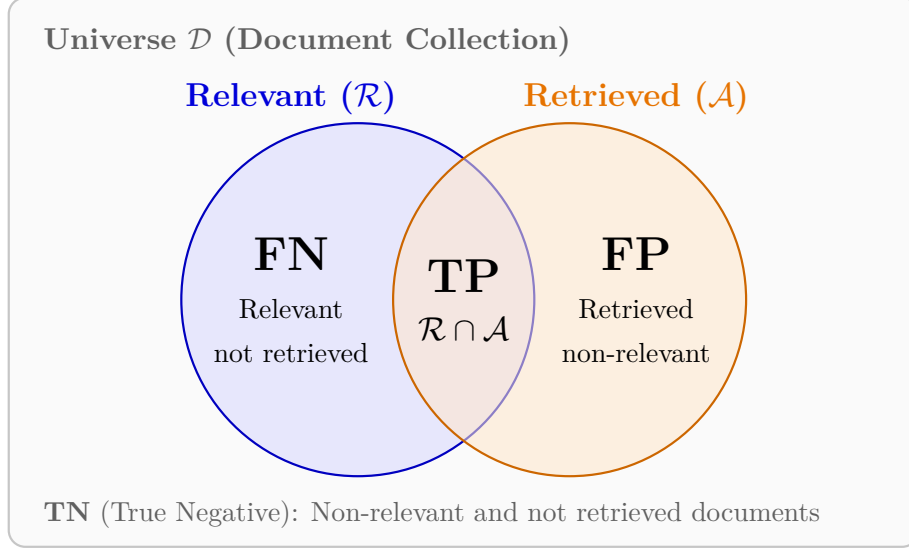


Figure 1.3: Set-theoretic Venn diagram of document retrieval and confusion regions.

### 1.6.1 Precision, Recall, and the Precision-Recall Trade-off

**Definition 1.6.1** (Precision ( $P$ )). The fraction of retrieved documents that are relevant:

$$P = \frac{TP}{TP + FP} = P(\text{relevant} \mid \text{retrieved}) \quad (1.1)$$

**Definition 1.6.2** (Recall ( $R$ )). The fraction of relevant documents in the collection that were successfully retrieved:

$$R = \frac{TP}{TP + FN} = P(\text{retrieved} \mid \text{relevant}) \quad (1.2)$$

An inherent **trade-off** governs Precision and Recall:

- Returning the entire document repository (Retrieved =  $\mathcal{D}$ ) guarantees  $R = 100\%$ , but drops precision near zero ( $P \approx 0$ ).
- Conversely, returning only the single most confident document can yield  $P = 100\%$ , at the expense of dismal recall.

### 1.6.2 The Harmonic Mean: F-Measure ( $F_1$ -Score)

To balance Precision and Recall into a single scalar metric that penalizes extreme disparity, the **F-Measure** calculates their **harmonic mean**:

$$F_1 = \frac{2}{\frac{1}{P} + \frac{1}{R}} = \frac{2 \cdot P \cdot R}{P + R} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad (1.3)$$

**Property 1.6.1** (Harmonic Mean Properties). *The harmonic mean exhibits key mathematical properties for evaluation:*

1. *It is always upper-bounded by both the geometric and arithmetic means:*

$$\frac{2}{\frac{1}{a} + \frac{1}{b}} \leq \sqrt{a \cdot b} \leq \frac{a + b}{2}$$

2. *When the two operands diverge significantly, the harmonic mean is heavily drawn toward the **minimum** rather than the arithmetic average.*

**Example 1.6.1** (Edge Behavior of F-Measure). Consider a degenerate retrieval system returning all documents:  $R = 1.0$  and  $P = 0.0001$ .

- The arithmetic mean yields an artificially inflated score:

$$\frac{1.0 + 0.0001}{2} \approx 0.50 \quad (50\%)$$

- The harmonic mean ( $F_1$ ) penalizes the system:

$$F_1 = \frac{2 \cdot 1.0 \cdot 0.0001}{1.0 + 0.0001} \approx 0.0002 \quad (0.02\%)$$

This illustrates why the harmonic mean is mathematically favored over the arithmetic mean.

### 1.6.3 The Weighted $F_\beta$ -Measure

When asymmetric priority is given to Precision versus Recall, a weighting parameter  $\alpha \in [0, 1]$  is introduced:

$$F = \frac{1}{\alpha \frac{1}{P} + (1 - \alpha) \frac{1}{R}} \quad (1.4)$$

Setting  $\alpha = \frac{1}{1+\beta^2}$  leads to the standard  $F_\beta$  formulation:

$$F_\beta = \frac{(1 + \beta^2) \cdot P \cdot R}{\beta^2 \cdot P + R} \quad (1.5)$$

- Values of  $\beta < 1$  (e.g.,  $\beta = 0.5$ ) emphasize **Precision** over Recall (favored in Web search).
- Values of  $\beta > 1$  (e.g.,  $\beta = 2.0$ ) emphasize **Recall** over Precision (crucial in legal discovery, patent search, and clinical meta-analyses).

## 1.7 Rank-Based Measures (Binary Relevance)

Modern search engines return ranked lists ordered by descending relevance scores. Furthermore, Web users rarely look beyond the first page of results ( $k = 10$  or  $k = 30$ ).

### 1.7.1 Precision@k ( $P@k$ ) and Recall@k ( $R@k$ )

**Definition 1.7.1** (Precision@k). Given a fixed rank cutoff  $k \in \mathbb{N}^+$ , Precision@k is the fraction of relevant documents within the top  $k$  positions:

$$P@k = \frac{1}{k} \sum_{i=1}^k \text{rel}_i \quad (1.6)$$

where  $\text{rel}_i \in \{0, 1\}$  represents the binary relevance of the document at rank  $i$ .

**Example 1.7.1** (Computing Precision@k). Consider a ranked list with binary relevance sequence for the top 5 positions  $\langle 1, 1, 0, 0, 1 \rangle$ :

- $P@1 = \frac{1}{1} = 1.00$
- $P@2 = \frac{2}{2} = 1.00$
- $P@3 = \frac{2}{3} \approx 0.66$
- $P@4 = \frac{2}{4} = 0.50$  (note that  $P@4 < P@3$ )
- $P@5 = \frac{3}{5} = 0.60$

### The Distortion of $P@k$ When Averaged Across Queries

While  $P@k$  is intuitive for estimating top-page result quality, it suffers from a critical methodological flaw when averaged across heterogeneous queries:

- If a specific query  $q_A$  possesses only  $R_A = 2$  relevant documents in the entire corpus, the theoretical maximum possible  $P@10(q_A)$  is capped at  $\frac{2}{10} = 0.20$ , even if a flawless engine places both relevant documents at ranks 1 and 2!
- Averaging raw  $P@10$  across queries penalizes the retrieval engine for the intrinsic scarcity of relevant documents for specific information needs, rather than algorithmic ranking errors.

Similarly, the complementary measure  $Recall@k$  ( $R@k$ ) evaluates the fraction of all relevant documents recovered within rank  $k$ :

$$Recall@k = \frac{1}{R} \sum_{i=1}^k rel_i \quad (1.7)$$

### 1.7.2 Average Precision (AP)

**Average Precision** (AP) summarizes the entire Precision-Recall curve into a single scalar score for a given query, evaluating instantaneous precision at each rank position where recall increases.

**Definition 1.7.2** (Average Precision). For a ranking of  $N$  documents corresponding to a query with  $R$  total relevant documents in the collection:

$$AP = \frac{1}{R} \sum_{k=1}^N P@k \cdot rel_k \quad (1.8)$$

**Property 1.7.1** (Geometric Interpretation and the Role of  $1/R$ ). *Average Precision carries a direct analytical meaning:*

1. *Each time a relevant document is encountered at rank  $k$ , recall increases by a discrete increment  $\Delta Recall = \frac{1}{R}$ . AP accumulates the rectangular areas  $P@k \cdot \Delta Recall$ , representing the **discrete numerical approximation of the Area Under the Precision-Recall Curve** (AUC-PR).*
2. *Normalizing by  $R$  (the total number of relevant documents in the corpus) ensures that relevant documents omitted from the top- $N$  candidate pool*

*implicitly contribute zero precision, heavily penalizing systems with incomplete recall.*

|                        | Search Engine Output Order |         |         |         |         |
|------------------------|----------------------------|---------|---------|---------|---------|
| Rank Position ( $k$ ): | $k = 1$                    | $k = 2$ | $k = 3$ | $k = 4$ | $k = 5$ |
| Document:              | Rel                        | NonRel  | Rel     | NonRel  | Rel     |
| Precision@ $k$ :       | 1.00                       | —       | 0.67    | —       | 0.60    |

Figure 1.4: Schematic representation of Average Precision calculation over 5 documents with  $R = 3$  total relevant items.

**Example 1.7.2** (Basic Calculation with  $R = 3$  Relevant Documents). Consider the ranking scenario in Figure 1.4 with  $R = 3$  total relevant documents appearing at ranks  $k_1 = 1$ ,  $k_2 = 3$ , and  $k_3 = 5$ :

$$P@1 = \frac{1}{1} = 1.00, \quad P@3 = \frac{2}{3} \approx 0.667, \quad P@5 = \frac{3}{5} = 0.60$$

The resulting Average Precision is:

$$AP = \frac{1}{3} \left( 1.00 + \frac{2}{3} + \frac{3}{5} \right) = \frac{1}{3} (2.267) \approx 0.756 \quad (75.6\%) \quad (1.9)$$

**Example 1.7.3** (Comparative Analysis Between Two Rankings). Consider a query associated with  $R = 6$  relevant documents in the collection. Two competing search algorithms output the following top-10 ranked results:

- **Ranking #1:** Relevant items concentrated early in the list:

$\langle \text{Rel}, \text{NonRel}, \text{Rel}, \text{Rel}, \text{Rel}, \text{Rel}, \text{NonRel}, \text{NonRel}, \text{NonRel}, \text{Rel} \rangle$

- **Ranking #2:** Relevant items delayed and scattered:

$\langle \text{NonRel}, \text{Rel}, \text{NonRel}, \text{NonRel}, \text{Rel}, \text{Rel}, \text{Rel}, \text{NonRel}, \text{Rel}, \text{Rel} \rangle$

Table 1.3 presents the step-by-step evolution of Precision and Recall across all 10 ranks.

| Rank ( $k$ )                  | Ranking #1 |        |             | Ranking #2  |        |             |
|-------------------------------|------------|--------|-------------|-------------|--------|-------------|
|                               | Item       | Recall | Precision   | Item        | Recall | Precision   |
| 1                             | <b>Rel</b> | 0.17   | <b>1.00</b> | NonRel      | 0.00   | —           |
| 2                             | NonRel     | 0.17   | —           | <b>Rel</b>  | 0.17   | <b>0.50</b> |
| 3                             | <b>Rel</b> | 0.33   | <b>0.67</b> | NonRel      | 0.17   | —           |
| 4                             | <b>Rel</b> | 0.50   | <b>0.75</b> | NonRel      | 0.17   | —           |
| 5                             | <b>Rel</b> | 0.67   | <b>0.80</b> | <b>Rel</b>  | 0.33   | <b>0.40</b> |
| 6                             | <b>Rel</b> | 0.83   | <b>0.83</b> | <b>Rel</b>  | 0.50   | <b>0.50</b> |
| 7                             | NonRel     | 0.83   | —           | <b>Rel</b>  | 0.67   | <b>0.57</b> |
| 8                             | NonRel     | 0.83   | —           | NonRel      | 0.67   | —           |
| 9                             | NonRel     | 0.83   | —           | <b>Rel</b>  | 0.83   | <b>0.56</b> |
| 10                            | <b>Rel</b> | 1.00   | <b>0.60</b> | <b>Rel</b>  | 1.00   | <b>0.60</b> |
| <b>Average Precision (AP)</b> |            |        | <b>0.78</b> | <b>0.52</b> |        |             |

Table 1.3: Detailed comparative evaluation of Average Precision between Ranking #1 and Ranking #2.

The resulting AP scores evaluate to:

$$AP(\text{Ranking \#1}) = \frac{1.00 + 0.67 + 0.75 + 0.80 + 0.83 + 0.60}{6} = \frac{4.65}{6} \approx \mathbf{0.78}$$

$$AP(\text{Ranking \#2}) = \frac{0.50 + 0.40 + 0.50 + 0.57 + 0.56 + 0.60}{6} = \frac{3.13}{6} \approx \mathbf{0.52}$$

Although both systems achieve identical overall recall ( $R = 100\%$  across 10 retrieved documents), Ranking #1 obtains a substantially higher score (0.78 vs 0.52) because it rewards front-loading relevant items.

### 1.7.3 Mean Average Precision (MAP)

To quantify overall effectiveness across a suite of  $|\mathcal{Q}|$  benchmark queries, we compute the arithmetic mean of their respective AP scores:

**Definition 1.7.3** (Mean Average Precision).

$$MAP = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} AP(q) \quad (1.10)$$

**Property 1.7.2** (Macro-Averaging vs. Micro-Averaging). *MAP formally implements a **Macro-Averaging** scheme: each query carries identical statistical weight  $1/|\mathcal{Q}|$ .*

- *In contrast, Micro-Averaging (pooling all instantaneous precisions across all queries and dividing by the total sum of relevant documents) would be overwhelmingly dominated by a handful of broad queries (e.g., "movies" or "news", which often match thousands of relevant documents), drowning out performance on specific long-tail queries.*
- *Macro-Averaging ensures that every individual information need receives equal representation in the final benchmark metric.*

**Example 1.7.4** (Computing MAP Over Two Queries). Consider an evaluation suite consisting of two distinct queries  $q_1$  and  $q_2$ :

- **Query 1 ( $R_1 = 5$  total relevant documents):** The engine returns 5 relevant items at ranks  $k \in \{1, 3, 6, 9, 10\}$  with precisions 1.00, 0.67, 0.50, 0.44, and 0.50:

$$AP(q_1) = \frac{1.00 + 0.67 + 0.50 + 0.44 + 0.50}{5} = \frac{3.11}{5} \approx \mathbf{0.62}$$

- **Query 2 ( $R_2 = 3$  total relevant documents):** The engine returns 3 relevant items at ranks  $k \in \{2, 5, 7\}$  with precisions 0.50, 0.40, and 0.43:

$$AP(q_2) = \frac{0.50 + 0.40 + 0.43}{3} = \frac{1.33}{3} \approx \mathbf{0.44}$$

The overall Mean Average Precision across both queries is:

$$MAP = \frac{AP(q_1) + AP(q_2)}{2} = \frac{0.62 + 0.44}{2} = \mathbf{0.53} \quad (1.11)$$

#### 1.7.4 Truncated Metrics: AP@K and MAP@K

In Web search and Recommender Systems, evaluating AP across an entire collection is impractical (the true total count  $R$  of relevant items is unknown) and disconnected from user browsing habits. Engines thus adopt rank-truncated formulations (**MAP@K**):

**Definition 1.7.4** (AP@K and MAP@K). Given a fixed rank cutoff  $K$  (typically  $K = 10$  or  $K = 20$ ), Average Precision at  $K$  for query  $q_i$  is defined as:

$$AP@K(q_i) = \frac{1}{K_i} \sum_{k=1}^K P@k(q_i) \cdot \text{rel}(q_i, k) \quad (1.12)$$

where the normalizing denominator  $K_i$  reflects the number of relevant items within the top  $K$ :

$$K_i = \sum_{k=1}^K \text{rel}(q_i, k) \quad (\text{or } K_i = \min(R_i, K)) \quad (1.13)$$

The global **MAP@K** across query suite  $\mathcal{Q}$  evaluates to:

$$\text{MAP@K}(\mathcal{Q}) = \frac{1}{|\mathcal{Q}|} \sum_{q_i \in \mathcal{Q}} \text{AP@K}(q_i) \quad (1.14)$$

This formulation resolves the fundamental flaw of raw  $P@K$ : by normalizing by  $K_i$  instead of a rigid  $K$ , a query with only 2 relevant documents that places both in the top 2 slots attains  $\text{AP@10} = 1.0$  rather than being artificially capped at 0.20.

### 1.7.5 Mean Reciprocal Rank (MRR)

In tasks such as known-item search, navigational queries (e.g., locating an institutional homepage), or factoid retrieval, the user is interested solely in finding the **first correct document**. User effort is inversely proportional to the rank of that first hit:

**Definition 1.7.5** (Reciprocal Rank and MRR). Let  $\text{rank}_1(q)$  denote the rank position of the first relevant document returned for query  $q$ . The **Reciprocal Rank** (RR) is:

$$\text{RR}(q) = \frac{1}{\text{rank}_1(q)} \quad (1.15)$$

The **Mean Reciprocal Rank** (MRR) across all queries is:

$$\text{MRR} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{1}{\text{rank}_1(q)} \quad (1.16)$$

If the first relevant hit appears at rank 1, the score is 1.0; if at rank 2, it drops to 0.5; if delayed to rank 10, it falls to 0.1.

## 1.8 Beyond Binary Relevance: Graded Relevance and NDCG

In practical Web and commercial applications, binary judgments prove inadequate. Consider the following real-world Google search query:

Query: "information retrieval unipi"

A commercial SERP returns pages exhibiting distinct qualitative levels of utility:

1. **Course DidaWiki (Instructor, CFU, syllabus, lecture notes):** The primary resource fully satisfying user search intent  $\implies$  **Highly Relevant** ( $r = 3$ ).
2. **Official UniPi Valutami Portal (Exam registration and administrative syllabus):** Authoritative institutional record but lacks study material  $\implies$  **Relevant** ( $r = 2$ ).
3. **Master's Degree PDF Brochure:** Contains only a brief mention of the course  $\implies$  **Marginally Relevant** ( $r = 1$ ).
4. **Unrelated articles or campus news:**  $\implies$  **Irrelevant** ( $r = 0$ ).

While all top three pages are pertinent, user satisfaction depends heavily on displaying the most informative page at rank 1. To capture this nuance, IR formalizes multi-grade relevance scales ( $r \in \{0, 1, 2, 3\}$  or  $\{0, \dots, 4\}$ ).

The **Discounted Cumulative Gain** (DCG) framework, introduced by Järvelin and Kekäläinen [8], rests on two assumptions:

1. Highly relevant documents are much more useful than marginally relevant ones.
2. The lower a document appears in the ranking, the less likely the user is to examine it; hence, gain must be discounted with descending rank.

### 1.8.1 Cumulative Gain (CG) and Discounted Cumulative Gain (DCG)

**Definition 1.8.1** (Cumulative Gain at Rank  $p$ ). Given a vector of graded relevance values  $\mathbf{r} = [r_1, r_2, \dots, r_p]$ , the Cumulative Gain is the unweighted sum:

$$CG_p = \sum_{i=1}^p r_i \quad (1.17)$$

$CG_p$  is position-insensitive: swapping a highly relevant document at rank 1 with one at rank 10 leaves  $CG_p$  unchanged.

**Definition 1.8.2** (Discounted Cumulative Gain (Standard Formulation)). Standard DCG applies a logarithmic discount based on rank  $i$ :

$$DCG_p = r_1 + \sum_{i=2}^p \frac{r_i}{\log_2(i)} \quad (1.18)$$

### Behavioral Meaning of the Logarithmic Discount

The discount factor  $\frac{1}{\log_2(i)}$  reflects the exponential decay of user visual attention under cascade reading models:

- At rank  $i = 4$ , the discount factor is  $\frac{1}{\log_2(4)} = \frac{1}{2}$ : perceived informative utility is cut in half relative to rank 1.
- At rank  $i = 8$ , the factor drops to  $\frac{1}{\log_2(8)} = \frac{1}{3}$ .
- At rank  $i = 16$ , the factor falls to  $\frac{1}{\log_2(16)} = \frac{1}{4}$ .

Commercial search engines (Google, Bing) frequently utilize the **exponential formulation**:

$$DCG_p^{\text{exp}} = \sum_{i=1}^p \frac{2^{r_i} - 1}{\log_2(i + 1)} \quad (1.19)$$

### Summation Index Clarification

Note the index shift in the logarithm denominator:

- In Equation (1.18),  $r_1$  is separated to prevent division by  $\log_2(1) = 0$ , while the summation starts at  $i = 2$  with  $\log_2(i)$ .
- In Equation (1.19), the summation starts directly at  $i = 1$  by taking  $\log_2(i + 1)$  ( $\log_2(1 + 1) = \log_2(2) = 1$ ).
- Furthermore, numerator  $2^{r_i} - 1$  non-linearly rewards top-tier relevance ( $r = 3 \implies 2^3 - 1 = 7$ , vs.  $r = 1 \implies 2^1 - 1 = 1$ ), severely penalizing any demotion of stellar results.

## 1.8.2 Normalized Discounted Cumulative Gain

Because different queries have varying numbers of relevant documents, raw  $DCG_p$  scores cannot be compared across queries. We normalize by the **Ideal DCG** ( $IDCG_p$ ), obtained by computing the DCG of the ideal ranking sorted by relevance descending:

$$NDCG_p = \frac{DCG_p}{IDCG_p} \in [0, 1] \quad (1.20)$$

### 1.8.3 Complete Numerical Walkthrough

Consider the following 10-document ranking, judged on a scale of 0–3:

$$\mathbf{r} = [3, 2, 3, 0, 0, 1, 2, 2, 3, 0]$$

The ideal sorted ranking is:

$$\mathbf{r}^* = [3, 3, 3, 2, 2, 2, 1, 0, 0, 0]$$

Table 1.4 summarizes the step-by-step progression across all 10 ranks.

| $i$ | $r_i$ | Discount             | Discounted Gain | $DCG@i$     | $r_i^*$ | Ideal Gain | $IDCG@i$     | $NDCG@i$    |
|-----|-------|----------------------|-----------------|-------------|---------|------------|--------------|-------------|
| 1   | 3     | 1.000                | 3.00            | <b>3.00</b> | 3       | 3.00       | <b>3.00</b>  | <b>1.00</b> |
| 2   | 2     | $\log_2(2) = 1.000$  | 2.00            | <b>5.00</b> | 3       | 3.00       | <b>6.00</b>  | <b>0.83</b> |
| 3   | 3     | $\log_2(3) = 1.585$  | 1.89            | <b>6.89</b> | 3       | 1.89       | <b>7.89</b>  | <b>0.87</b> |
| 4   | 0     | $\log_2(4) = 2.000$  | 0.00            | <b>6.89</b> | 2       | 1.00       | <b>8.89</b>  | <b>0.76</b> |
| 5   | 0     | $\log_2(5) = 2.322$  | 0.00            | <b>6.89</b> | 2       | 0.86       | <b>9.75</b>  | <b>0.71</b> |
| 6   | 1     | $\log_2(6) = 2.585$  | 0.39            | <b>7.28</b> | 2       | 0.77       | <b>10.52</b> | <b>0.69</b> |
| 7   | 2     | $\log_2(7) = 2.807$  | 0.71            | <b>7.99</b> | 1       | 0.36       | <b>10.88</b> | <b>0.73</b> |
| 8   | 2     | $\log_2(8) = 3.000$  | 0.67            | <b>8.66</b> | 0       | 0.00       | <b>10.88</b> | <b>0.80</b> |
| 9   | 3     | $\log_2(9) = 3.170$  | 0.95            | <b>9.61</b> | 0       | 0.00       | <b>10.88</b> | <b>0.88</b> |
| 10  | 0     | $\log_2(10) = 3.322$ | 0.00            | <b>9.61</b> | 0       | 0.00       | <b>10.88</b> | <b>0.88</b> |

Table 1.4: Detailed trace of  $DCG@i$ ,  $IDCG@i$ , and  $NDCG@i$  computation for ranks 1 through 10.

### 1.8.4 Comparing Alternative Ranking Functions

Consider 4 documents  $\{d_1, d_2, d_3, d_4\}$  with graded assessments:

$$r(d_1) = 0, \quad r(d_2) = 1, \quad r(d_3) = 2, \quad r(d_4) = 2$$

- **Ground Truth (GT):** Ordering  $\langle d_4, d_3, d_2, d_1 \rangle$  with relevance vector  $[2, 2, 1, 0]$ :

$$IDCG_4 = 2 + \frac{2}{\log_2(2)} + \frac{1}{\log_2(3)} + 0 = 2 + 2 + 0.6309 = 4.6309$$

$$NDCG_{GT} = 1.00$$

- **Ranking Function 1 ( $RF_1$ ):** Ordering  $\langle d_3, d_4, d_2, d_1 \rangle$  with relevance  $[2, 2, 1, 0]$ . Because the first two documents share identical relevance ( $r = 2$ ), their permutation incurs no positional penalty:

$$DCG_{RF_1} = 2 + \frac{2}{\log_2(2)} + \frac{1}{\log_2(3)} = 4.6309$$

$$NDCG_{RF_1} = \frac{4.6309}{4.6309} = 1.00$$

- **Ranking Function 2 ( $RF_2$ ):** Ordering  $\langle d_3, d_2, d_4, d_1 \rangle$  with relevance  $[2, 1, 2, 0]$ . Here,  $d_2$  ( $r = 1$ ) erroneously outranks  $d_4$  ( $r = 2$ ):

$$DCG_{RF_2} = 2 + \frac{1}{\log_2(2)} + \frac{2}{\log_2(3)} + 0 = 2 + 1 + 1.2619 = 4.2619$$

$$NDCG_{RF_2} = \frac{DCG_{RF_2}}{IDCG_4} = \frac{4.2619}{4.6309} \approx 0.9203$$

This illustrates how NDCG mathematically penalizes rank inversions in direct proportion to the relevance difference.

# Chapter 2

## Algorithmic Efficiency on Modern CPU Architectures

### 2.1 The Role of Efficiency in Information Retrieval

The development of information processing systems unfolds along two complementary dimensions that are often in tension with one another:

1. **Effectiveness:** maximizes end-user satisfaction, focusing on the semantic quality of results, ranking, and natural language processing, typically using high-level abstraction tools (Python, Machine Learning frameworks).
2. **Efficiency:** minimizes infrastructural operating costs, designing algorithms optimized down to the last CPU clock cycle and the last bit of occupied memory.

No single professional figure is required to master both extremes; however, those who develop at a high level must understand the physical constraints of the underlying hardware to avoid designing impractical solutions, while those who optimize at a low level must preserve the structure and semantics of the data so as not to compromise its effectiveness.

#### 2.1.1 The Three Pillars of Efficiency in IR Systems

**User experience.** In web-scale systems, a query response must be delivered within an infinitesimal fraction of a second. Users perceive any delay beyond a few tenths of a second: if results are slow or irrelevant, they abandon

the service in favor of a competitor. Algorithmic efficiency does not only serve to respond faster, but allows more sophisticated and effective ranking and filtering algorithms to run *within the same predetermined time budget* (typically a hard ceiling of 100–200 ms).

**Operational scalability.** Modern systems handle billions of documents, multimedia items, queries, and logs generated every second. The number of machines required is approximately:

$$\text{Number of machines} = \left\lceil \frac{\text{Total data volume}}{\text{Single-node operating capacity}} \right\rceil.$$

If the data volume doubles, a well-designed system requires at most twice as many machines (linear scalability); an inefficient algorithm can instead require superlinear cost growth (e.g., quadrupling servers), which is unsustainable at scale. To meet required latencies, active data must reside in RAM and avoid disk access: this requires algorithms that are efficient not only in time but also in space, via compression techniques. Furthermore, since data is replicated across multiple machines for fault tolerance, every memory inefficiency is amplified by the infrastructure’s replication factor.

**Environmental and economic sustainability.** Reducing the number of physical machines directly cuts data center energy consumption. High-density servers dissipate large amounts of heat, and much of the energy is consumed by cooling systems; for this reason many large-scale data centers are located in cold regions or near bodies of water (*free cooling*). A marginal computational saving, even just 3%, generates significant economic savings and reduces the carbon footprint.

## 2.2 The Memory Hierarchy and Locality Principles

### 2.2.1 The Latency Gap Between CPU and Memory

In the classical theoretical computational model (abstract RAM machine), access to any memory location has constant unit cost,  $\mathcal{O}(1)$ . On real physical machines, the CPU’s arithmetic units operate at speeds enormously higher than main memory (DRAM), and there exists a hierarchy of intermediate levels (*caches*) that approximates fast access for frequently used data.

| Hardware level        | Typical latency     | CPU cycles          | Typical capacity     |
|-----------------------|---------------------|---------------------|----------------------|
| CPU instruction (ALU) | 0.5–5 ns            | 1–2                 | internal registers   |
| L1I / L1D Cache       | 0.5 ns              | $\approx 1$         | 32 KB per core       |
| L2 Cache              | 1.5 ns              | $\approx 3$ –4      | 256 KB (up to 2 MB)  |
| L3 Cache (LLC)        | 5.0 ns              | $\approx 10$ –15    | 16 MB (up to 64 MB+) |
| Main RAM (DRAM)       | 80.0 ns             | $\approx 160$ –200+ | 64 GB+               |
| Disk / Storage        | 500 000 ns (0.5 ms) | millions            | 8 TB+                |

Table 2.1: Quantitative comparison of latency and capacity across memory hierarchy levels.

**Definition 2.2.1** (*Memory-Bound Algorithm*). An algorithm that accesses main RAM in a disordered fashion pays  $\approx 80$  ns per access, an interval during which the CPU could execute hundreds of instructions. Such an algorithm is called **memory-bound**: execution time is dominated by waiting for data transfer from DRAM, and the CPU spends most of its time stalled, making the optimization of arithmetic operation count irrelevant.

## 2.2.2 Physical On-Chip Architecture

The L1 cache is integrated close to the decode unit and execution units; to avoid bus conflicts it is physically split into **L1I** (*Instruction Cache*, holding fetched machine code for execution) and **L1D** (*Data Cache*, holding operands for **load/store** instructions). The L2 cache is integrated within the single core and acts as a low-latency buffer; the L3 cache is external to individual cores but internal to the package, and is shared among all cores of the machine.

### The Cache Line as the Minimum Transfer Unit

RAM and the cache hierarchy never transfer single bytes or isolated individual variables: the minimum hardware transfer unit is the **cache line**, typically  $B = 64$  bytes. If a program needs a 64-bit integer (8 bytes) located at address  $A$ , the hardware is forced to load the entire aligned block

$$[\lfloor A/64 \rfloor \times 64, \lfloor A/64 \rfloor \times 64 + 63].$$

If the program reads that single value and then jumps to a random address, the remaining 56 bytes (87.5% of the memory traffic) are wasted.

### 2.2.3 Temporal and Spatial Locality

A program effectively exploits the memory hierarchy only if it respects at least one of the two forms of locality:

- **Temporal locality:** when a datum accessed at time  $t$  is likely to be accessed again shortly thereafter (e.g., loop counters, frequently reused variables).
- **Spatial locality:** when accessing an address makes it likely that nearby addresses will be accessed soon after (e.g., sequential array traversal), which hardware exploits through the cache line and the **prefetcher** (a unit that detects linear access patterns and speculatively loads upcoming cache lines before they are explicitly requested).

**Definition 2.2.2** (Sequential Access vs. *Pointer Chasing*). A **sequential** access pattern visits contiguous memory addresses in increasing order, without isolated loops or disjoint sub-cycles. **Pointer chasing** instead follows a chain of pointers whose target addresses are unpredictable in advance and typically scattered across memory, defeating spatial locality and prefetching.

Both functions below compute the sum of all elements of the vector,  $\sum_{i=0}^{N-1} v[i] = \frac{(N-1)N}{2}$ .

Listing 2.1: Linear sequential scan.

```
1 pub fn sum(v: &[usize]) -> usize {
2     let mut sum = 0;
3     for &value in v.iter() {
4         sum += value;
5     }
6     sum
7 }
```

Listing 2.2: Scan via random jumps (*pointer chasing*).

```
1 pub fn sum_by_jumping(v: &[usize]) -> usize {
2     let mut sum = 0;
3     let mut k = 0;
4     for _ in 0..v.len() {
5         sum += v[k];
6         k = v[k]; // the value read becomes the index
7                   for the next iteration
8     }
9     sum
10 }
```

In the theoretical RAM model both functions execute  $N$  iterations with one addition per step: they have identical asymptotic complexity  $\mathcal{O}(N)$ . On real hardware, measuring the average time per element as array size varies, marked divergences emerge instead.

| $N$        | Size    | Cache level | Time per elem. |           | Slowdown |
|------------|---------|-------------|----------------|-----------|----------|
|            |         |             | sum()          | jumping() |          |
| 256        | 2 KiB   | L1 (32 KB)  | 0.1 ns         | 1.3 ns    | 13×      |
| 8 192      | 64 KiB  | L2 (256 KB) | 0.1 ns         | 2.5 ns    | 25×      |
| 262 144    | 2 MiB   | L3 (16 MB)  | 0.2 ns         | 13.4 ns   | 67×      |
| 16 777 216 | 128 MiB | RAM (DRAM)  | 0.4 ns         | 84.9 ns   | 212×     |

Table 2.2: Average time per element for `sum()` and `sum_by_jumping()` as array size varies.

## 2.2.4 Analysis of Physical Causes

**Why does `sum()` only vary between 0.1 and 0.4 ns?** With 64-bit `usize` integers (8 bytes), a 64-byte cache line holds  $64/8 = 8$  contiguous elements: there is full spatial locality. Accessing a new line incurs an initial miss, but the following 7 accesses occur at near-zero cost from cache, and the stream prefetcher loads subsequent lines ahead of execution. Denoting by  $T_{\text{comp}}$  the computation time and by  $T_{\text{load}}$  the amortized transfer time, the observed cost per element follows

$$\text{Observed latency} \approx \max(T_{\text{comp}}, T_{\text{amortized\_load}}),$$

almost entirely masked by computation; the increase from 0.1 to 0.4 ns at 128 MiB is due solely to saturation of the bus bandwidth toward RAM.

**Why does `sum_by_jumping()` reach 84.9 ns?** Since the permutation is random, the address  $k_{t+1} = v[k_t]$  points to an arbitrary location: the probability that it falls in the same cache line is negligible, so for every 8-byte element read the remaining 56 transferred bytes are discarded (total absence of spatial locality). With no linear or constant stride, prefetching is inhibited. Moreover, the CPU must know the index of the next read before it can issue it: this is a **serial dependency** that prevents any overlap. When the array size exceeds L3 capacity (128 MiB  $\gg$  16 MB), every access becomes a global cache miss to DRAM:

$$\text{Time per element} \approx \text{physical DRAM access latency} \approx 80\text{--}85 \text{ ns},$$

a value that matches exactly the 84.9 ns measured.

**Example 2.2.1** (Amortizing the Cost of *Pointer Chasing* by Exploiting the Cache Line). If the jump occurs between 64-byte blocks (8 64-bit integers), but all elements of the block are summed before the next jump, the cost of a cache miss to RAM ( $\approx 85$  ns) is paid only once every 8 addition operations:

```

1 pub fn sum_by_jumping_cache_lines(v: &[usize]) -> usize
2     {
3         let mut sum = 0;
4         let mut k = 0;
5         let num_blocks = v.len() / 8;
6         for _ in 0..num_blocks {
7             let base = k * 8;
8             // fully exploits the cache line already loaded
9             // into memory
10            sum += v[base] + v[base + 1] + v[base + 2] + v[
11                base + 3]
12                + v[base + 4] + v[base + 5] + v[base + 6];
13            k = v[base + 7] % num_blocks; // pointer to the
14            next jump
15        }
16        sum
17    }

```

The latency is amortized over 8 elements, bringing the expected average time per value to

$$\text{Average time per value} \approx \frac{85 \text{ ns}}{8} \approx 10.6 \text{ ns},$$

an intermediate result between linear scanning and element-by-element random jumping.

**Remark 2.2.1** (Eliminating the Residual Slowdown of `sum()`). The slight slowdown from 0.1 to 0.4 ns observed in `sum()` occurs because  $T_{\text{comp}} \approx 0.1$  ns is lower than the time needed to feed the data stream from RAM for very large vectors (memory-bound regime). By increasing the computational complexity of the loop body (e.g. multiplications or non-trivial functions on each element) one obtains  $T_{\text{comp}} \gg T_{\text{load\_RAM}}$ : the relation  $\max(T_{\text{comp}}, T_{\text{load}})$  becomes constantly equal to  $T_{\text{comp}}$ , the regime becomes *CPU-bound*, and the time curve becomes perfectly flat as  $N$  varies.

## 2.3 Forms of Parallelism and Pipeline Architecture

Modern CPUs exploit multiple forms of parallelism: **multi-threading/multi-core** (multiple independent physical cores on the same processor), **pipelining** (splitting instruction execution into stages operating simultaneously on adjacent instructions), and **superscalar** execution (multiple compute units within the same core to process several independent instructions per cycle).

### Embarrassingly Parallel Workloads in Search Engines

Search engines do not need to synchronize multiple cores to answer a single query: they adopt **trivial parallelism** (*embarrassingly parallel*), in which, given thousands of simultaneous users, each thread on a single core serves a specific query in complete isolation. With no shared state, lock contention and cache coherence issues are eliminated.

### 2.3.1 Anatomy of the Five-Stage Pipeline

Pipelining decomposes instruction execution into independent sequential stages, which in a standard architecture are five:

1. **IF** (*Instruction Fetch*): fetching the machine-code instruction from memory (L1I) via the *program counter*.
2. **ID** (*Instruction Decode*): decoding the instruction into micro-operations ( $\mu$ -ops) and reading source registers.
3. **EX** (*Execute*): performing the arithmetic-logic computation in the ALUs or computing the effective memory address.
4. **MEM**: access to the memory subsystem (L1D) for load or store instructions (LOAD/STORE).
5. **WB** (*Write-Back*): writing the result to architectural registers and retiring the instruction.

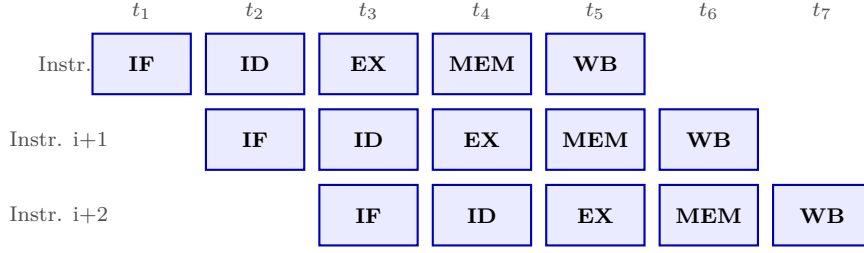


Figure 2.1: Overlap of the five pipeline stages across consecutive instructions.

### 2.3.2 Latency, Throughput, and Pipeline Depth

Let  $k$  be the number of pipeline stages and  $\tau$  the clock cycle period. The **latency**  $L$  is the total time required for a single instruction to traverse the entire pipeline,  $L = k \cdot \tau$ : in modern CPUs the latency for a simple operation typically ranges between 10 and 20 clock cycles. The **throughput**  $W$  is the number of instructions completed per unit time, and at steady state  $W = 1/\tau$  instructions per second, since one instruction exits the pipeline every cycle. The period  $\tau$  is dictated by the delay of the slowest stage in the circuit chain,

$$\tau \geq \max_{1 \leq i \leq k} (t_{\text{stage}_i}).$$

To raise the clock frequency (reduce  $\tau$ ), designers increase pipeline depth  $k$  (up to 15–20 or more stages in modern CPUs), implementing each stage with simpler, faster circuitry; conversely, increasing  $k$  inflates the penalty in case of a forced pipeline flush.

## 2.4 Superscalar Execution, Interleaving, and Structural Hazards

A processor is called **superscalar** when it has the architectural ability to issue and complete more than one instruction per clock cycle ( $IPC > 1$ , *Instructions Per Cycle*) within the same core, integrating multiple arithmetic-logic units (integer ALUs, multipliers, shift units, load/store units) served by parallel issue ports (*execution ports*). Real architectures such as Intel Haswell feature, for instance, a 56-entry decode queue, a 192-entry reorder buffer, a unified 60-entry reservation station, and 8 parallel execution ports, each specialized for classes of operations (integer ALUs, load/store, vector units, branch, etc.).

### 2.4.1 The Arithmetic *Interleaving* Experiment

Four Rust functions operating on a vector of 64-bit integers `&[u64]` are benchmarked, accumulating an increasing number of independent operations per iteration.

Listing 2.3: From a single reduction to four independent reductions per iteration.

```
1 pub fn do_stuff(v: &[u64]) -> u64 {
2     let mut prod = 1;
3     for &value in v.iter() {
4         prod *= value;
5     }
6     prod
7 }
8
9 pub fn do_more_stuff(v: &[u64]) -> (u64, u64) {
10     let mut prod = 1;
11     let mut sum = 0;
12     for &value in v.iter() {
13         prod *= value;
14         sum += value;
15     }
16     (prod, sum)
17 }
18
19 pub fn do_more_stuff_2(v: &[u64]) -> (u64, u64, u64) {
20     let mut prod = 1;
21     let mut sum = 0;
22     let mut xor = 0;
23     for &value in v.iter() {
24         prod *= value;
25         sum += value;
26         xor ^= value;
27     }
28     (prod, sum, xor)
29 }
30
31 #[inline]
32 pub fn do_more_stuff_3(v: &[u64]) -> (u64, u64, u64, u64)
33     ) {
34     let mut prod = 1;
35     let mut sum = 0;
36     let mut xor = 0;
```

```

36     let mut or = 0;
37     for &value in v.iter() {
38         prod *= value;
39         sum += value;
40         xor ^= value;
41         or |= value;
42     }
43     (prod, sum, xor, or)
44 }

```

| Function                         | Time per iteration |
|----------------------------------|--------------------|
| do_stuff() (1 operation)         | 0.4 ns             |
| do_more_stuff() (2 operations)   | 0.4 ns             |
| do_more_stuff_2() (3 operations) | 0.4 ns             |
| do_more_stuff_3() (4 operations) | 0.5 ns             |

Table 2.3: Time per iteration as the number of accumulated independent reductions grows.

## 2.4.2 Instruction Overlap and Port Saturation

From one to three operations the time remains unchanged at 0.4 ns because the accumulator variables have no mutual dependencies:  $\text{prod}_{t+1} = \text{prod}_t \cdot v[i]$ ,  $\text{sum}_{t+1} = \text{sum}_t + v[i]$ ,  $\text{xor}_{t+1} = \text{xor}_t \oplus v[i]$ . The superscalar core simultaneously dispatches the multiplication, addition, and XOR to three distinct execution ports: the three operations complete in the same clock cycle, and the extra work is performed at zero cost.

With the fourth operation (`or`) the time rises to 0.5 ns rather than doubling to 0.8 ns: this is a **structural hazard**, since the physical units available for parallel arithmetic-logic computation in a single cycle are saturated (only three instructions of that type can be issued simultaneously), and one of the four operations must wait. However the *out-of-order* CPU does not process iterations as isolated compartments: it implicitly unrolls the instruction window, grouping them into blocks of three, absorbing the excess operation into subsequent cycles and saturating the residual capacity of the ports. The real overhead thus amounts to a marginal fraction of a cycle ( $\Delta t = +0.1$  ns) rather than a full additional cycle.

## 2.5 Data Hazards and Serial Dependencies

**Definition 2.5.1** (Data Hazard). A **data hazard** occurs when an instruction needs an operand that is not yet available because it is still being computed by a preceding instruction. It corresponds to a **RAW** (*Read-After-Write*) data dependency:

$$\text{Inst}_A : R_1 \leftarrow R_2 + R_3 \quad \Longrightarrow \quad \text{Inst}_B : R_4 \leftarrow R_1 \times R_5,$$

where  $\text{Inst}_B$  must stall in the pipeline until  $\text{Inst}_A$  completes its computation.

Consider code that performs the same number of operations per iteration as the previous `do_more_stuff_3`, but chains the data in a tight dependency:

Listing 2.4: Serial RAW dependency chain across four operations.

```
1 pub fn do_more_stuff_4(v: &[u64]) -> (u64, u64, u64) {
2     let mut prod = 1;
3     let mut sum = 0;
4     let mut xor = 0;
5     for &value in v.iter() {
6         sum += value; // 1. compute sum
7         prod *= sum; // 2. tightly depends on sum
8         xor ^= prod; // 3. tightly depends on prod
9         sum += xor; // 4. depends on xor and
10                // overwrites sum for the next iteration!
11     }
12     (prod, sum, xor)
13 }
```

Because of the RAW chain, the execution units cannot start any computation in parallel:

$$\text{Iteration latency} = t_{\text{add}} + t_{\text{mul}} + t_{\text{xor}} + t_{\text{add\_feedback}}.$$

Since the last operation updates `sum` and the first instruction of the next iteration requires the new value of `sum`, the *out-of-order* CPU cannot start iteration  $t + 1$  early: the superscalar units remain unused. Experimentally, against the 0.5 ns of `do_more_stuff_3` (independent operations), `do_more_stuff_4` measures 1.7 ns per iteration ( $\approx 3.5\times$  slower), despite performing the same number of arithmetic operations.

### 2.5.1 Algorithmic Solution: *Interleaving* Independent Vectors

When running this computation on two distinct vectors `V` and `West`, the sequential approach (calling `do_more_stuff_4` on `V` and then on `West`) requires

total time  $2T$ . If instead a single function processes both vectors within the same iteration, alternating the two independent chains:

Listing 2.5: Fusion (interleaving) of two mutually independent RAW chains.

```

1  for i in 0..n {
2      // chain for V (occupies the ports at cycle k)
3      sum_v += v[i];
4      prod_v *= sum_v;
5      xor_v ^= prod_v;
6      sum_v += xor_v;
7      // chain for West, completely independent of V!
8      sum_w += west[i];
9      prod_w *= sum_w;
10     xor_w ^= prod_w;
11     sum_w += xor_w;
12 }
```

the measured execution time is identical to that of the single vector *V* alone ( $\approx T$ , rather than  $2T$ ). Operations on *West* do not depend on those on *V*: while *V*'s chain is waiting for ALU results, the free compute units process *West*'s chain, filling the dead time imposed by data dependencies at zero extra cost.

## 2.6 Control Hazards, Speculative Execution, and Branch Prediction

**Definition 2.6.1** (Control Hazard). A **control hazard** occurs when the CPU cannot determine which instruction to fetch next (*instruction fetch*) due to the presence of a conditional branch (*if*, *branch*).

The CPU cannot stop at every condition: it therefore adopts **speculative execution**, guessing via a *branch predictor* which branch will be taken and speculatively fetching subsequent instructions to feed the pipeline. If the outcome of the condition turns out to disagree with the prediction, all speculative work completed in the intermediate stages must be discarded, with a full pipeline flush and reload from the correct branch:

Misprediction penalty = 15–20 clock cycles.

### 2.6.1 The Random Condition Paradox

A conditional filter is introduced into code with a dependency chain, operating on an array of random integers uniformly distributed in  $[0, 99]$ :

Listing 2.6: Conditional branch on random values: the CPU cannot predict the outcome.

```
1 pub fn do_more_stuff_5(v: &[u64]) -> (u64, u64, u64) {
2     let mut prod = 1;
3     let mut sum = 0;
4     let mut xor = 0;
5     for &value in v.iter() {
6         if value > 50 { // true with p = 0.5 on random
7             // values in [0, 99]
8             sum += value;
9             prod *= sum;
10            xor ^= prod;
11            sum += xor;
12        }
13    }
14    (prod, sum, xor)
}
```

Since the condition is true on average 50% of the time, one would intuitively expect a time roughly halved compared to `do_more_stuff_4` ( $1.7/2 \approx 0.85$  ns). The measured time instead rises to 3.5 ns, more than twice as slow as the version without the condition.

#### The Real Cost Is Not Computation but Misprediction

With random values the branch predictor's accuracy is equivalent to a coin flip,  $P(\text{hit}) = P(\text{miss}) = 0.50$ . The average cost per iteration is then:

$$\begin{aligned}\text{Average cost} &\approx 0.5 \times T_{\text{body}} + P(\text{miss}) \times \text{Flush penalty} \\ &\approx 0.5 \times 0.85 \text{ ns} + 0.5 \times 5.0 \text{ ns} \approx 2.9\text{--}3.5 \text{ ns},\end{aligned}$$

having assumed a penalty of 15–20 cycles ( $\approx 4\text{--}5$  ns on a 4 GHz CPU). The cost of the algorithm is no longer dictated by arithmetic operations, but is dominated by the time lost flushing and refilling the pipeline.

### 2.6.2 Experimental Verification: Data Ordering and Periodic Patterns

To demonstrate that the slowdown derives exclusively from branch predictor behavior, the same identical code is run while only changing the arrangement of data in the array.

| Configuration                        | Time / iter. | Branch predictor behavior                      |
|--------------------------------------|--------------|------------------------------------------------|
| do_more_stuff_4() (no if)            | 1.7 ns       | —                                              |
| do_more_stuff_5() (random data)      | 3.5 ns       | Unpredictable branch, miss rate $\approx 50\%$ |
| do_more_stuff_5() (sorted data)      | 1.2 ns       | Near-perfect prediction, single miss           |
| do_more_stuff_5() (alternating data) | 1.1 ns       | Periodic sequence, miss rate $\approx 0\%$     |

Table 2.4: Time per iteration of `do_more_stuff_5()` as data arrangement varies.

With **sorted data** (all elements  $\leq 50$  at the start of the array and all elements  $> 50$  at the end) the predictor assumes the branch is not taken for the whole first half and taken for the second: a single misprediction occurs at the transition point, and once flush costs are removed the time drops to 1.2 ns, lower than the 1.7 ns of the version without the condition since half the iterations do not execute the `if` body. With **alternating data** following a periodic pattern (*false, true, false, true, ...*), predictors integrate historical pattern tables (*branch history shift register*): once the periodicity is identified, the predictor anticipates the outcome at every step without errors, eliminating stalls.

### 2.6.3 *Branchless* Techniques and the Implicit Jump of `for` Loops

If data cannot be sorted or predicted, the only solution is to physically remove the conditional branch (*branchless code*), replacing it with algebraic expressions or conditional assembly instructions (`CMOV`). For example, instead of

```

1 if value > 50 {
2     sum += value;
3 }
```

the logical guard is mapped onto a scalar binary value (0 or 1), always performing the arithmetic operation:

```

1 let condition = (value > 50) as u64; // 1 if true, 0 if
   false
2 sum += value * condition;           // adds 'value' if
   true, adds '0' if false
```

The program always performs the mathematical computation, but the instruction flow is linear: there are no branches, and mispredictions are eliminated.

**Remark 2.6.1** (Why the `for` Loop Branch Is Not a Problem). Every `for` loop ends at the machine level with a conditional jump back to the start of the block. This branch does not degrade performance because it is highly asymmetric: for  $N$  iterations, the loop-continuation condition is true  $N - 1$  consecutive times and false only upon exit. The branch predictor immediately learns the pattern (“branch taken”) and commits a single misprediction at the end of the loop, with a completely negligible impact on overall time.

## 2.7 Synthesis: From Hardware Constraints to Algorithmic Choices

The phenomena analyzed in this chapter – memory hierarchy and locality, pipelining, superscalar execution, *data hazards*, and *control hazards* – are not negligible implementation details, but physical constraints that determine the actual performance behavior of large-scale Information Retrieval algorithms. In particular:

- data structures (posting lists, inverted indexes) should be designed to favor sequential, contiguous access, in order to exploit spatial locality and hardware prefetching;
- when possible, it is beneficial to express multiple independent reductions within the same loop to saturate superscalar execution ports at marginal cost;
- tight (RAW) dependency chains between accumulator variables should be identified and, where possible, decoupled or interleaved with independent computations from other data streams;
- unpredictable conditional branches on data-dependent conditions should be replaced with branchless techniques whenever the misprediction cost outweighs the extra computation.

# Chapter 3

## Natural Language Processing and Text Preprocessing

### 3.1 Recap: Graded Relevance Evaluation Metrics (CG, DCG, NDCG)

In real-world Information Retrieval systems, the binary relevance assumption (a document is either relevant or non-relevant) fails to capture qualitative nuances across retrieved results. Relevance judgments are instead recorded on discrete graded scales (e.g., Likert scales from 0 to 3 or 0 to 4, reflecting increasing degrees of pertinence).

Evaluating a ranked list at cutoff depth  $p$  relies on two core principles:

1. **Gain:** placing highly relevant documents at top positions of the Search Engine Results Page (SERP) significantly boosts user satisfaction.
2. **Positional Discounting:** relevant documents placed at lower ranks yield diminishing perceived utility, since human attention decays monotonically down the list.

#### 3.1.1 Cumulative Gain (CG) and Discounted Cumulative Gain (DCG)

Given a graded relevance vector  $\mathbf{r} = [r_1, r_2, \dots, r_p]$  for the top- $p$  documents retrieved for a query  $q$ :

**Definition 3.1.1** (Cumulative Gain at Rank  $p$ ). **Cumulative Gain** ( $CG_p$ )

sums relevance scores linearly up to rank  $p$ :

$$CG_p = \sum_{i=1}^p r_i. \quad (3.1)$$

*Structural limitation:*  $CG_p$  is position-insensitive: swapping an essential document ( $r = 3$ ) between position 1 and position 10 leaves  $CG$  unchanged.

To model user attention decay across rank positions  $i$ , logarithmic discount functions are applied:

**Definition 3.1.2** (Discounted Cumulative Gain at Rank  $p$ ). Under the standard formulation and the exponential gain variant widespread in commercial web search:

$$DCG_p = r_1 + \sum_{i=2}^p \frac{r_i}{\log_2(i)}, \quad (3.2)$$

$$DCG_p^{\text{exp}} = \sum_{i=1}^p \frac{2^{r_i} - 1}{\log_2(i + 1)}. \quad (3.3)$$

The exponential formulation (3.3) heavily rewards placing top-relevance documents at the very highest ranks.

### 3.1.2 Normalized Discounted Cumulative Gain (NDCG) and Query Normalization

Raw DCG values are strictly **query-dependent**. If query  $q_A$  has 100 relevant documents in the corpus, its theoretical maximum DCG will be huge; if query  $q_B$  has only one relevant document, its maximum DCG will be small. Averaging raw DCG across different queries is therefore mathematically invalid.

To enable benchmark comparability (*macro-averaging*), **Ideal DCG** ( $IDCG_p$ ) is defined as the DCG of the ideal ranking obtained by sorting all documents by descending relevance up to depth  $p$ :

$$NDCG_p = \frac{DCG_p}{IDCG_p} \in [0, 1]. \quad (3.4)$$

An  $NDCG_p$  of 1.0 guarantees that the algorithm achieved the best possible ordering up to depth  $p$ .

## 3.2 Fundamental Properties of Natural Language

### 3.2.1 Distinctive Characteristics of Human Language

Automated processing of natural language faces structural bottlenecks inherent to human linguistic systems:

- **Thousands of discrete symbols:** lexicons consist of hundreds of thousands of distinct words and morphological stems, expanding with domain-specific terms and neologisms.
- **Complex and recursive syntax:** sentences form hierarchical dependency trees with long-distance syntactic bindings.
- **Predominantly compositional semantics:** overall sentence meaning derives from combining constituent units. Notable exceptions include:
  - *Collocations:* conventional recurrent phrases (e.g., “*make the bed*”, “*skip class*”).
  - *Idiomatic expressions:* figurative meanings not deductible from individual constituents (e.g., “*break a leg*”, “*out of the frying pan into the fire*”).
- **Pervasive ambiguity:**
  - *Part-of-Speech (POS) Ambiguity:* identical words serving distinct grammatical roles depending on context (e.g., “*beat*”: noun in “*Listen to this cool beat*”, verb in “*I’ll beat you at checkers*”).
  - *Word Sense Ambiguity (Polysemy and Homonymy):* multiple meanings for a single surface form (e.g., “*interest*”: intellectual curiosity vs financial interest rate).
  - *Syntactic Ambiguity:* multiple valid parse trees. Groucho Marx’s famous joke illustrates this: “*I shot an elephant in my pajamas*” (the prepositional phrase can attach to the subject or the object).
- **Shared world knowledge dependence:** comprehension relies on extra-textual knowledge. “*I’m faster than Lewis Hamilton*” requires knowing Hamilton is an F1 driver; “*I was on a bike with a helmet*” implies wearing the helmet, whereas “*I was on a bike with an electric motor*” implies the motor is mounted on the bike.

- **Continuous temporal evolution:** semantic drift (e.g., “*in the cloud*”) and rapid emergence of technological and cultural neologisms (“*to google*”, “*crowdfunded*”, “*super spreader*”).

### 3.2.2 Zipf’s Law and the Principle of Least Effort

Term occurrence frequencies in natural corpora follow an empirical power-law distribution formulated by linguist George Kingsley Zipf:

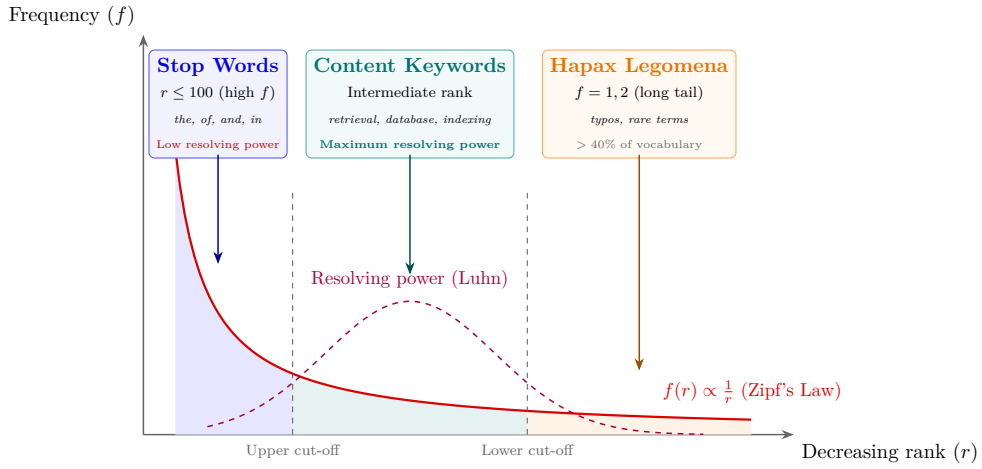


Figure 3.1: Hyperbolic Zipf distribution and vocabulary resolving power (Luhn’s hypothesis): partitioning into high-frequency stop words ( $r \leq 100$ ), optimal medium-rank index terms, and the long tail of hapax legomena.

**Definition 3.2.1** (Zipf’s Law). Sorting corpus terms by descending frequency, the term at rank  $r$  has empirical frequency  $f$  inversely proportional to  $r$ :

$$f \propto \frac{1}{r} \implies f \cdot r \approx C, \quad (3.5)$$

where  $C$  is a corpus-specific scaling constant.

In Mandelbrot’s generalized formulation (*Zipf-Mandelbrot law*):

$$f(r) \propto \frac{1}{(r + b)^a}, \quad (3.6)$$

with empirical parameters typically fitted around  $a \approx 1$  and  $b \approx 2.7$ .

### Principle of Least Effort

Zipf theorized this distribution emerges from an economic equilibrium between competing cognitive drivers:

1. **Speaker's Effort:** minimizes articulatory expense using a compact vocabulary of short, frequent, polysemous words.
2. **Hearer's Effort:** favors an expansive vocabulary of rare, distinct, non-ambiguous words to minimize semantic disambiguation cost.

The evolutionary balance produces the observed hyperbolic curve.

Two verifiable empirical consequences follow:

- In the standard benchmark corpus *Reuters 21578* ( $\approx 500\,000$  total word occurrences), just **313 distinct words** account for over 50% of all token occurrences.
- A word's number of distinct dictionary meanings  $m$  is inversely proportional to the square root of its frequency:  $m \propto 1/\sqrt{f}$ .

### 3.2.3 Rare Terms and Stop Words

Zipf's distribution exposes a clear vocabulary bifurcation:

1. **Stop Words:** ultra-high frequency words at minimum ranks ( $r \leq 100$ ), mostly function words, articles, prepositions, and conjunctions (*the, of, a, in, to*). They carry negligible topical information for discriminating documents. However, aggressive stripping harms phrase search (e.g., "*to be or not to be*") and sentiment classification.
2. **Rare Terms (Hapax Legomena):** words occurring only once or twice across the collection. They constitute 40%–50% of unique vocabulary terms, encompassing spelling errors (*typos*), technical slugs, or rare keywords with very high discriminatory power.

## 3.3 Text Acquisition and Preprocessing Pipeline

### 3.3.1 Web Text Extraction

Search engine pipelines start by ingesting raw web data:

1. **Document Fetching (Network I/O):** Python's `urllib.request` issues HTTP requests to pull raw HTML payloads:

Listing 3.1: Downloading web markup using `urllib`.

```
1 from urllib import request
2
3 url = "http://example.org/page.html"
4 response = request.urlopen(url)
5 html = response.read().decode('utf8')
```

2. **HTML Stripping and Text Extraction:** HTML formatting markup, layout attributes, and scripts represent non-informative noise. Dedicated parsers like `BeautifulSoup` isolate textual content:

Listing 3.2: Isolating clean textual content using `BeautifulSoup`.

```
1 from bs4 import BeautifulSoup
2
3 soup = BeautifulSoup(html, "html5lib")
4 # Global unstructured text extraction
5 raw_text = soup.get_text()
6
7 # Structured paragraph-level extraction
8 paragraphs = [' '.join(s.findAll(text=True)) for s
               in soup.findAll('p')]
```

### 3.3.2 Segmentation and Tokenization

**Sentence Splitting.** Sentence boundaries cannot be determined simply by period punctuation (`.`). Periods appear in acronyms (*U.S.A.*), honorific abbreviations (*Dr.*, *St.*), decimal numbers (*8.30*), and ellipses. NLTK's `sent_tokenize()` employs a trained classifier to distinguish contextual punctuation roles.

**Word Tokenization.** Splits the character stream into discrete lexical units (*tokens*):

Listing 3.3: Standard word tokenization with NLTK.

```
1 from nltk import word_tokenize
2
3 tokens = word_tokenize("The President of the United
   States of America")
```

```
4 # Output: ['The', 'President', 'of', 'the', 'United', 'States', 'of', 'America']
```

Specialized tokenizers handle specific domains: `TweetTokenizer` for hash-tags, handles, and emojis, or `RegexTokenizer` for regular-expression-driven extraction.

### 3.3.3 Morphological Normalization: Stemming vs. Lemmatization

Morphological normalization collapses inflected and derived surface variants into a canonical representation, shrinking index vocabulary dimensionality.

#### Stemming

Heuristic rule-based string transformation (stripping or substituting affixes):

- **Properties:** highly computationally efficient, operates on individual words without requiring dictionaries or syntax trees.
- **Pitfalls:** frequently produces non-existent root stems (*overstemming* and *understemming*).
- **Example (Porter Stemmer):**
  - "cars" → "car" (correct plural stripping).
  - "was" → "wa" (the algorithm mistakenly strips the trailing *s* as an inflectional plural suffix).

#### Lemmatization

Principled morphological process that maps inflected surface tokens to their authentic dictionary lemma:

- **Properties:** leverages *Part-of-Speech (POS) Tagging* and comprehensive lexical resources (such as *WordNet*).
- **Example (WordNetLemmatizer):**
  - Default call without POS (treated as noun): `lemmatize("was")` outputs "wa".
  - Specifying proper POS (verb): `lemmatize("was", pos='v')` returns the dictionary lemma "be".

## 3.4 Representation Models: Bag of Words and N-Grams

### 3.4.1 The Bag of Words (BoW) Model

The Bag of Words model discards syntactic ordering and grammatical dependencies, abstracting each document into an unordered multiset (or set) of independent tokens:

Listing 3.4: Basic set-based text representation.

```
1 from nltk import word_tokenize
2
3 text = 'the president of the united states of america'
4 tokens = word_tokenize(text) # Length: 8 tokens
5 bow_set = set(tokens)        # Cardinality: 6 unique
   elements
6 # bow_set: {'america', 'of', 'president', 'states', 'the', 'united'}
```

### Structural Limitations of Bag of Words

1. **Loss of syntactic order and compositional semantics:** Consider two statements with completely opposite real-world meanings:

- $t_1$ : “I won, and thus you lose.”
- $t_2$ : “I lose, and thus you won.”

Under set-based BoW representation:

$$\text{set}(t_1) \equiv \text{set}(t_2) = \{\text{'and'}, \text{'lose'}, \text{'thus'}, \text{'won'}, \text{'you'}\}. \quad (3.7)$$

To the retrieval system, these opposite assertions are completely identical, causing **semantic collapse**.

2. **Loss of multiplicity (in binary set versions):** fails to record how many times a key term appears.

### 3.4.2 N-Gram Models

To recover local context without bearing full syntactic parsing overhead, vocabularies are enriched with ***n*-grams** (contiguous sequences of  $n$  items).

## Word N-grams

Extracts sequences of  $n$  consecutive words ( $n = 2$  for bigrams,  $n = 3$  for trigrams), retaining local word order:

- Applied to the victory/loss example ( $n = 2$ ):

$$\text{ngrams}(t_1, 2) = \{\text{'I won'}, \text{'won and'}, \text{'and thus'}, \text{'thus you'}, \text{'you lose'}\}, \quad (3.8)$$

$$\text{ngrams}(t_2, 2) = \{\text{'I lose'}, \text{'lose and'}, \text{'and thus'}, \text{'thus you'}, \text{'you won'}\}. \quad (3.9)$$

Now the representations differ ( $\text{ngrams}(t_1, 2) \neq \text{ngrams}(t_2, 2)$ ), preserving who won and who lost.

## Character N-grams

Extracts substrings of  $n$  consecutive characters within words (commonly  $n = 3$  trigrams):

- **Role in IR:** provides resilience against typing errors (*typo-tolerance*) and captures morphological similarities.
- **Example with typographical error** ("rainbow" vs. "rainbaw"):

$$C_3(\text{"rainbow"}) = \{\text{'r-a-i'}, \text{'a-i-n'}, \text{'i-n-b'}, \text{'n-b-o'}, \text{'b-o-w'}\}, \quad (3.10)$$

$$C_3(\text{"rainbaw"}) = \{\text{'r-a-i'}, \text{'a-i-n'}, \text{'i-n-b'}, \text{'n-b-a'}, \text{'b-a-w'}\}. \quad (3.11)$$

The intersection is non-empty:

$$C_3(\text{"rainbow"}) \cap C_3(\text{"rainbaw"}) = \{\text{'r-a-i'}, \text{'a-i-n'}, \text{'i-n-b'}\}. \quad (3.12)$$

Sharing 3 out of 5 trigrams (60%), the search engine identifies high proximity and suggests “*Did you mean: rainbow?*”.

# Chapter 4

## Retrieval Models, Scoring, and TF-IDF Weighting

### 4.1 Evolutionary Paradigms of Information Retrieval

The technological evolution of Information Retrieval spans four major architectural eras:

1. **Statistical Era (1960s – Mid-1990s):** relevance functions relied purely on lexical statistics within documents (*tf*) and collection frequency (*idf*). This framework was vulnerable to *keyword stuffing*, where artificial repetition of keywords trivially boosted rankings.
2. **Structural Era and PageRank (1998):** Sergey Brin and Larry Page addressed the collapse of pure lexical ranking by exploiting Web graph topology. Inbound hyperlinks function as authority endorsements from source pages. Manipulating the global link graph is exponentially harder than altering on-page text.
3. **Machine Learning and Learning-to-Rank Era (2000s):** relevance was modeled through supervised learners fusing hundreds of disparate features: text matching signals, link graph reputation, and user interaction signals (click-through rates, query reformulation, dwell time).
4. **Neural Era and Dense Encoders (2018 – Present):** Transformers enabled deep contextual linguistic understanding. Documents and queries are projected into dense metric embedding spaces or weighted via learned sparse expansions.

## 4.2 Boolean Retrieval vs. Ranked Retrieval

The distinction between exact propositional matching and continuous scored ranking defines modern retrieval systems:

| Property                  | Boolean Retrieval                                                         | Ranked Retrieval                                                                 |
|---------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Matching Criterion</b> | Exact, binary propositional truth<br>( $d \models q \implies \{0, 1\}$ ). | Continuous real-valued relevance score:<br>$\text{Score}(q, d) \in \mathbb{R}$ . |
| <b>Query Formulation</b>  | Constrained Boolean expressions (AND, OR, NOT).                           | Free natural language queries ( <i>free-text query</i> ).                        |
| <b>Returned Output</b>    | Unordered set of satisfying documents.                                    | Monotonically sorted ranking by estimated relevance score.                       |
| <b>User Experience</b>    | <i>Feast or Famine</i> pathology.                                         | Immediate delivery of top- $k$ results (typically $k \approx 10$ ).              |
| <b>Target Audience</b>    | Expert searchers (information specialists, patent lawyers).               | General web users.                                                               |

Table 4.1: Systematic comparison between Boolean and Ranked Retrieval models.

### The “Feast or Famine” Pathology

Unranked Boolean search causes two undesirable extremes for regular users:

- **Famine:** using conjunctive AND constraints excessively restricts matches. Missing even one keyword (e.g., due to synonymy) completely excludes a document, often returning zero results.
- **Feast:** disjunctive OR operators relax constraints too aggressively, returning tens of thousands of unranked documents without quality guidance.

## 4.3 Set-Based Proximity Measures: The Jaccard Coefficient

A foundational approach to grading document relevance without vector algebra employs the Jaccard set overlap coefficient:

**Definition 4.3.1** (Jaccard Similarity Coefficient). Given two term sets  $A$  and  $B$ , the Jaccard similarity is the ratio of their intersection cardinality over their union cardinality:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}. \quad (4.1)$$

*Properties:* bounded within  $[0, 1]$ , where  $J(A, A) = 1$  and  $J(A, B) = 0 \iff A \cap B = \emptyset$ .

### 4.3.1 Worked Example and the Document Length Paradox

Consider query  $q$  and two candidate documents:

- **Query  $q$ :** “*ides of march*”  
 $Q = \{\text{ides, of, march}\}, |Q| = 3$ .
- **Document  $d_1$ :** “*caesar died in march*”  
 $D_1 = \{\text{caesar, died, in, march}\}, |D_1| = 4$ .
- **Document  $d_2$ :** “*the long march*”  
 $D_2 = \{\text{the, long, march}\}, |D_2| = 3$ .

**Analytical Evaluation:**

$$Q \cap D_1 = \{\text{march}\} \implies |Q \cap D_1| = 1, \quad (4.2)$$

$$J(q, d_1) = \frac{1}{3 + 4 - 1} = \frac{1}{6} \approx 0.166. \quad (4.3)$$

$$Q \cap D_2 = \{\text{march}\} \implies |Q \cap D_2| = 1, \quad (4.4)$$

$$J(q, d_2) = \frac{1}{3 + 3 - 1} = \frac{1}{5} = 0.200. \quad (4.5)$$

### The Jaccard Relevance Paradox

Because  $J(q, d_2) = 0.200 > J(q, d_1) = 0.166$ , Jaccard ranks  $d_2$  ahead of  $d_1$ . However, historically and semantically,  $d_1$  describes the assassination of Julius Caesar (the actual Ides of March), while  $d_2$  describes an unrelated historical retreat (the Chinese Long March). Document  $d_2$  obtains a higher score simply because it is shorter, inflating the denominator union  $|A \cup B|$  less.

## 4.3.2 Structural Flaws of Jaccard Similarity for IR

1. **Ignores Term Frequency ( $tf$ ):** whether a query term occurs once or 50 times in a document, its set intersection count remains exactly 1.
2. **Ignores Collection Term Rarity ( $df$ ):** pervasive stopwords like “*of*” contribute the exact same weight as ultra-discriminating keywords like “*ides*”.
3. **Unjust Length Penalization:** comprehensive and relevant documents with broader vocabularies suffer artificial denominator inflation, collapsing their score.

## 4.4 From Incidence Matrices to Count Matrices

Moving beyond discrete sets, document modeling naturally transitions into linear algebra.

### 4.4.1 Binary Incidence Matrix

Given vocabulary  $V = \{t_1, \dots, t_{|V|}\}$  and corpus  $D = \{d_1, \dots, d_N\}$ , the incidence matrix  $M \in \{0, 1\}^{|V| \times N}$  records binary term presence:

$$M_{t,d} = \begin{cases} 1 & \text{if } t \in d, \\ 0 & \text{otherwise.} \end{cases} \quad (4.6)$$

### 4.4.2 Term–Document Count Matrix

Reflecting topical emphasis through occurrence frequencies yields the count matrix  $C \in \mathbb{N}^{|V| \times N}$ , where  $C_{t,d} = \text{count}(t, d) = tf_{t,d}$ .

| Term      | Antony & Cleo. | Julius Caesar | The Tempest | Hamlet | Othello | Macbeth |
|-----------|----------------|---------------|-------------|--------|---------|---------|
| Antony    | 157            | 73            | 0           | 0      | 0       | 0       |
| Brutus    | 4              | 157           | 0           | 1      | 0       | 0       |
| Caesar    | 232            | 227           | 0           | 2      | 1       | 1       |
| Calpurnia | 0              | 10            | 0           | 0      | 0       | 0       |
| Cleopatra | 57             | 0             | 0           | 0      | 0       | 0       |
| mercy     | 2              | 0             | 3           | 5      | 5       | 1       |
| worser    | 2              | 0             | 1           | 1      | 1       | 0       |

Table 4.2: Term–document frequency count matrix across selected Shakespearean plays.

## 4.5 Term Frequency Weighting: Log-Frequency Weighting

### 4.5.1 Why Raw Counts are Inadequate

Relevance perceived by users does not scale linearly with raw term frequencies:

- A document mentioning a word 10 times is likely more relevant than one mentioning it once, but it is certainly **not 10 times more relevant**.
- Additional occurrences offer diminishing marginal returns of topical information.

### 4.5.2 Logarithmic Dampening of $tf$

To dampen the influence of large raw counts, sublinear logarithmic weighting is applied:

$$w_{t,d} = \begin{cases} 1 + \log_{10}(tf_{t,d}) & \text{if } tf_{t,d} > 0, \\ 0 & \text{otherwise.} \end{cases} \quad (4.7)$$

**Weight Progression:**

- $tf = 0 \implies w = 0$
- $tf = 1 \implies w = 1 + \log_{10}(1) = 1.0$
- $tf = 2 \implies w = 1 + \log_{10}(2) \approx 1.301$
- $tf = 10 \implies w = 1 + \log_{10}(10) = 2.0$
- $tf = 100 \implies w = 1 + \log_{10}(100) = 3.0$

- $tf = 1000 \implies w = 1 + \log_{10}(1000) = 4.0$

In continuous optimization settings,  $\log(1 + tf_{t,d})$  is frequently used, preserving zero-origin continuity ( $\log(1 + 0) = 0$ ).

Scoring purely on log-dampened term frequencies computes:

$$\text{Score}(q, d) = \sum_{t \in q \cap d} (1 + \log_{10}(tf_{t,d})). \quad (4.8)$$

## 4.6 Document Frequency ( $df$ ) and Inverse Document Frequency ( $idf$ )

### 4.6.1 Semantic Informativeness and Corpus Rarity

Local within-document frequency ( $tf$ ) does not reflect collection-wide specificity:

- Pervasive terms present in nearly all documents (e.g., *increase*, *high*, conjunctions) offer minimal discriminative power.
- Highly localized terms appearing in few documents (e.g., *Calpurnia*, *arachnocentric*) carry concentrated topical signal.

### 4.6.2 Mathematical Formulation of IDF

Let  $N = |D|$  denote total collection size, and let  $df_t$  denote the document frequency of term  $t$ . **Inverse Document Frequency (IDF)** is defined as:

$$idf_t = \log_{10} \left( \frac{N}{df_t} \right). \quad (4.9)$$

The logarithm compresses  $N/df_t$ , ensuring weights remain numerically well-scaled.

### 4.6.3 Worked Collection Example ( $N = 1\,000\,000$ )

Consider a collection of one million documents:

| Term ( $t$ ) | $df_t$    | Ratio $N/df_t$ | $idf_t = \log_{10}(N/df_t)$ |
|--------------|-----------|----------------|-----------------------------|
| Calpurnia    | 1         | $10^6$         | 6.0                         |
| animal       | 100       | $10^4$         | 4.0                         |
| sunday       | 1 000     | $10^3$         | 3.0                         |
| fly          | 10 000    | $10^2$         | 2.0                         |
| under        | 100 000   | $10^1$         | 1.0                         |
| the          | 1 000 000 | 1              | 0.0                         |

Table 4.3: IDF scale across varying Document Frequencies on  $N = 10^6$ .

#### Core Properties of IDF

1. **Automatic stopwords neutralization:** when a word appears everywhere ( $df_t = N$ ),  $idf_t = \log_{10}(1) = 0$ . The word contributes zero weight, naturally filtering stop words.
2. **Offline Query-Independent Precomputation:**  $idf_t$  depends solely on corpus statistics and **not on the incoming query**. All  $idf_t$  values are computed offline at indexing time and stored in the vocabulary dictionary for  $\mathcal{O}(1)$  online retrieval.

#### 4.6.4 IDF Impact on Ranking: Single-Term vs. Multi-Term Queries

**For single-term queries ( $|q| = 1$ ).** IDF has zero effect on the relative ranking among matching documents.

*Proof.* Let  $q = \{t\}$ . The IDF-weighted score for any matching document  $d$  containing  $t$  is:

$$\text{Score}(q, d) = \log_{10} \left( \frac{N}{df_t} \right). \quad (4.10)$$

Since this factor is identical for all matching documents, it acts as a constant multiplier. The relative ranking is determined entirely by  $tf_{t,d}$ .  $\square$

**For multi-term queries ( $|q| \geq 2$ ).** IDF becomes the **critical reweighting factor**: in a query like “*capricious person*”, the common word “*person*” (low  $idf$ ) is suppressed, while the rare keyword “*capricious*” (high  $idf$ ) drives the ranking.

### 4.6.5 Collection Frequency ( $cf_t$ ) vs. Document Frequency ( $df_t$ )

- **Collection Frequency ( $cf_t$ ):** total token occurrences across the entire collection:  $cf_t = \sum_d tf_{t,d}$ .
- **Document Frequency ( $df_t$ ):** number of unique documents containing  $t$  at least once.

**Example 4.6.1** (Comparison between “insurance” and “try”). In the lecture benchmark:

- “try”:  $cf = 10\,422$ , spread across  $df = 8\,760$  documents.
- “insurance”:  $cf = 10\,440$ , concentrated in only  $df = 3\,997$  documents.

Despite nearly identical total occurrences ( $cf$ ), “insurance” is far more topically focused and discriminative, receiving a substantially higher  $idf$ .

## 4.7 The Full TF-IDF Weighting Scheme

### 4.7.1 Mathematical Formulation

Combining local within-document saliency ( $tf$ ) with global corpus rarity ( $idf$ ) produces composite **tf-idf** weights:

$$w_{t,d} = \text{tf-idf}_{t,d} = wf_{t,d} \times idf_t, \quad (4.11)$$

where:

$$wf_{t,d} = \begin{cases} 1 + \log_{10}(tf_{t,d}) & \text{if } tf_{t,d} > 0, \\ 0 & \text{otherwise,} \end{cases} \quad idf_t = \log_{10} \left( \frac{N}{df_t} \right). \quad (4.12)$$

The preliminary matching score for an unweighted query  $q$  against document  $d$  is:

$$\text{Score}(q, d) = \sum_{t \in q \cap d} \text{tf-idf}_{t,d}. \quad (4.13)$$

### 4.7.2 Chalkboard Worked Case Study

**Initial setup:**  $N = 806\,791$ , Query: “car auto insurance best”.

| Term ( $t$ )     | $tf$ Doc 1 | $tf$ Doc 2 | $tf$ Doc 3 | $df_t$ | $idf_t = \log_{10}(N/df_t)$ |
|------------------|------------|------------|------------|--------|-----------------------------|
| <b>car</b>       | 27         | 4          | 24         | 18 165 | 1.65                        |
| <b>auto</b>      | 3          | 33         | 0          | 6 723  | 2.08                        |
| <b>insurance</b> | 0          | 33         | 29         | 19 241 | 1.62                        |
| <b>best</b>      | 14         | 0          | 17         | 25 235 | 1.50                        |

Table 4.4: Chalkboard exercise data: term frequencies and collection statistics.

**Case A: Raw Frequency Weighting** ( $tf_{t,d} \times idf_t$ ). For **Doc 1**:

- car:  $27 \times 1.65 = 44.55$
- auto:  $3 \times 2.08 = 6.24$
- insurance:  $0 \times 1.62 = 0.0$
- best:  $14 \times 1.50 = 21.0$

For **Doc 2**:

- car:  $4 \times 1.65 = 6.60$
- auto:  $33 \times 2.08 = 68.64$
- insurance:  $33 \times 1.62 = 53.46$
- best:  $0 \times 1.50 = 0.0$

For **Doc 3**:

- car:  $24 \times 1.65 = 39.60$
- auto:  $0 \times 2.08 = 0.0$
- insurance:  $29 \times 1.62 = 46.98$
- best:  $17 \times 1.50 = 25.50$

**Case B: Log-Dampened Weighting**  $\log_{10}(1 + tf_{t,d}) \times idf_t$ . Applying logarithmic dampening on **Doc 1**:

- car:  $\log_{10}(1 + 27) = \log_{10}(28) \approx 1.447 \implies 1.447 \times 1.65 \approx \mathbf{2.39}$
- auto:  $\log_{10}(1 + 3) = \log_{10}(4) \approx 0.602 \implies 0.602 \times 2.08 \approx \mathbf{1.25}$
- insurance:  $\log_{10}(1 + 0) = 0 \implies 0 \times 1.62 = \mathbf{0.0}$
- best:  $\log_{10}(1 + 14) = \log_{10}(15) \approx 1.176 \implies 1.176 \times 1.50 \approx \mathbf{1.76}$

Logarithmic scaling prevents massive raw counts from overpowering global informativeness conveyed by IDF.

### 4.7.3 The Weighted Term–Document Matrix

Replacing raw frequencies with real-valued  $\text{tf-idf}_{t,d}$  weights embeds the corpus into a real-valued matrix:

$$W \in \mathbb{R}^{|V| \times N}. \quad (4.14)$$

Each document is now mapped to a continuous vector  $\vec{d} \in \mathbb{R}^{|V|}$ , establishing the geometric foundation of the Vector Space Model.

# Chapter 5

## The Vector Space Model (VSM), SMART Notation, and Okapi BM25

### 5.1 Foundations of the Vector Space Model (VSM)

In the **Vector Space Model** (VSM), documents and queries are modeled as continuous geometric vectors residing in a real-valued finite-dimensional space:

$$\text{Dimensionality of the space} = |V|, \quad (5.1)$$

where each orthogonal axis corresponds to a distinct vocabulary term from  $V$ .

Both user queries  $q$  and candidate documents  $d$  are represented as multi-dimensional vectors:

$$\vec{v}(q) \in \mathbb{R}^{|V|}, \quad \vec{v}(d) \in \mathbb{R}^{|V|}. \quad (5.2)$$

#### 5.1.1 Extreme Sparsity of Document Vectors

In production web-scale environments, vocabulary cardinality easily reaches  $|V| \approx 10^6$ . An ordinary document contains between a few dozen and a few thousand distinct words:

$$\frac{|\{i \mid v_i(d) \neq 0\}|}{|V|} \leq \frac{10^3}{10^6} = 0.001 = 0.1\%. \quad (5.3)$$

Over **99.9%** of all vector dimensions are identically zero. Storage and arithmetic operations are therefore implemented strictly using sparse vector structures and inverted index posting lists.

## 5.2 Why Euclidean Distance Fails: The Document Duplication Experiment

The intuitive geometric distance metric in real coordinate spaces is Euclidean ( $L_2$ ) distance:

$$\text{dist}_{L_2}(\vec{q}, \vec{d}) = \|\vec{q} - \vec{d}\|_2 = \sqrt{\sum_{i=1}^{|V|} (q_i - d_i)^2}. \quad (5.4)$$

However, Euclidean distance is unsuited for Information Retrieval due to its sensitivity to extrinsic document length.

### 5.2.1 The Thought Experiment: Concatenated Duplicate Document ( $d' = d \circ d$ )

Consider a document  $d$  and duplicate it into  $d'$ , where every sentence is repeated twice:

1. **Semantic Content:**  $d'$  and  $d$  are identical regarding topic and relative term proportions.
2. **Vector Magnitude:** term frequencies double ( $tf_{t,d'} = 2 \cdot tf_{t,d}$ ), doubling vector norm ( $\|\vec{d'}\|_2 \approx 2\|\vec{d}\|_2$ ).
3. **Euclidean Distance Explosion:** the Euclidean distance between a short query  $\vec{q}$  and the long duplicate  $\vec{d'}$  grows substantially larger:

$$\|\vec{q} - \vec{d'}\|_2 \gg \|\vec{q} - \vec{d}\|_2. \quad (5.5)$$

Sorting by increasing Euclidean distance severely and unfairly penalizes longer documents.

4. **Unchanged Angular Direction:** both vectors  $\vec{d}$  and  $\vec{d'}$  lie **on the exact same ray** originating from the coordinate origin. The enclosed angle  $\theta$  between them is strictly  $0^\circ$ .

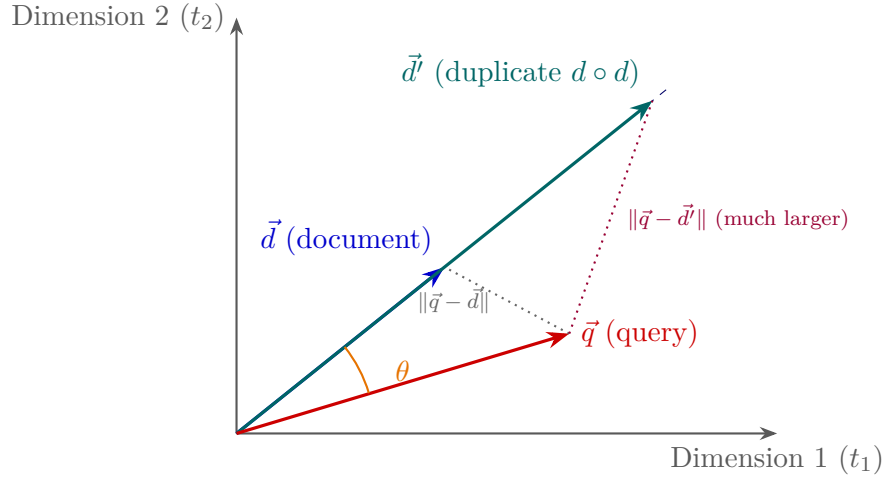


Figure 5.1: Euclidean distance failure:  $d$  and  $d'$  share identical direction ( $\theta = 0^\circ$ ), but  $d'$  has a much larger Euclidean distance to  $q$ .

#### Direction vs. Magnitude

Topical relevance must remain invariant to document verbosity. Relevance is captured by the **angular direction** of vectors, not the Euclidean distance between their endpoints.

### 5.3 From Angles to Cosine Similarity and $L_2$ Normalization

Sorting documents by increasing angle  $\theta$  (smaller angle  $\implies$  higher topical overlap) is mathematically equivalent to sorting by **decreasing cosine** of  $\theta$ , since the cosine function is monotonically decreasing over  $[0, \pi/2]$ :

$$\theta = 0^\circ \implies \cos(0^\circ) = 1 \quad (\text{maximum similarity, collinear vectors}), \quad (5.6)$$

$$\theta = 90^\circ \implies \cos(90^\circ) = 0 \quad (\text{orthogonal vectors, zero shared terms}). \quad (5.7)$$

Because TF-IDF weights are non-negative, vector representations are confined to the first positive hyper-octant:  $\cos(\theta) \in [0, 1]$ .

#### 5.3.1 Analytical Derivation of Cosine Similarity

From the geometric definition of the vector dot product:

$$\vec{q} \cdot \vec{d} = \|\vec{q}\|_2 \|\vec{d}\|_2 \cos(\theta), \quad (5.8)$$

where the Euclidean ( $L_2$ ) norm is:

$$\|\vec{x}\|_2 = \sqrt{\sum_{i=1}^{|V|} x_i^2}. \quad (5.9)$$

Solving for  $\cos(\theta)$  yields the canonical formula for **Cosine Similarity**:

$$\text{Sim}_{\cos}(\vec{q}, \vec{d}) = \cos(\theta) = \frac{\vec{q} \cdot \vec{d}}{\|\vec{q}\|_2 \|\vec{d}\|_2} = \frac{\sum_{t \in q \cap d} q_t d_t}{\sqrt{\sum_{t \in q} q_t^2} \sqrt{\sum_{t \in d} d_t^2}}. \quad (5.10)$$

### 5.3.2 $L_2$ Length Normalization onto the Unit Hypersphere

Dividere un vettore per la sua norma euclidea proietta il vettore sulla superficie dell'ipersfera unitaria:

$$\vec{v}_{\text{norm}} = \frac{\vec{v}}{\|\vec{v}\|_2} \implies \|\vec{v}_{\text{norm}}\|_2 = 1. \quad (5.11)$$

When all document vectors are pre-normalized to unit length, Cosine Similarity simplifies to a pure dot product:

$$\cos(\vec{q}, \vec{d}) = \vec{q}_{\text{norm}} \cdot \vec{d}_{\text{norm}} = \sum_{t \in q \cap d} q_{\text{norm},t} \cdot d_{\text{norm},t}. \quad (5.12)$$

### 5.3.3 Computational Decoupling: Offline vs. Online

- **Offline (Indexing):** the Euclidean length  $\|\vec{d}\|_2 = \sqrt{\sum_t d_t^2}$  depends exclusively on the document. It is calculated once during indexing and stored in the document lookup dictionary.
- **Online (Query Processing):** at query time,  $\|\vec{q}\|_2$  is computed. When traversing inverted lists, the engine accumulates the dot product across shared terms  $t \in q \cap d$  and normalizes the score:

$$\text{Score}(q, d) = \frac{1}{\|\vec{q}\|_2} \sum_{t \in q \cap d} \frac{q_t d_t}{\|\vec{d}\|_2}. \quad (5.13)$$

## 5.4 Worked Case Study: Stylistic Comparison Across Three Novels

The lecture analyzed lexical proximity across three English classics:

1. **SaS:** *Sense and Sensibility* (Jane Austen, 1811)
2. **PaP:** *Pride and Prejudice* (Jane Austen, 1813)
3. **WH:** *Wuthering Heights* (Emily Brontë, 1847)

The goal is to determine whether VSM geometry captures higher stylistic proximity between works by the same author than across distinct authors.

#### 5.4.1 Step 1: Raw Frequency Count Matrix ( $tf$ )

| Term             | SaS ( $d_1$ ) | PaP ( $d_2$ ) | WH ( $d_3$ ) |
|------------------|---------------|---------------|--------------|
| <b>affection</b> | 115           | 58            | 20           |
| <b>jealous</b>   | 10            | 7             | 11           |
| <b>gossip</b>    | 2             | 0             | 6            |
| <b>wuthering</b> | 0             | 0             | 38           |

Table 5.1: Raw term frequencies across the three novels.

#### 5.4.2 Step 2: Logarithmic Term Frequency Weighting ( $1 + \log_{10}(tf)$ )

Applying  $w = 1 + \log_{10}(tf)$  if  $tf > 0$ , and 0 otherwise:

- **SaS:**

- affection:  $1 + \log_{10}(115) \approx 1 + 2.0607 = 3.06$
- jealous:  $1 + \log_{10}(10) = 1 + 1.0 = 2.00$
- gossip:  $1 + \log_{10}(2) \approx 1 + 0.3010 = 1.30$
- wuthering: 0.00

- **PaP:**

- affection:  $1 + \log_{10}(58) \approx 1 + 1.7634 = 2.76$
- jealous:  $1 + \log_{10}(7) \approx 1 + 0.8451 = 1.85$
- gossip: 0.00
- wuthering: 0.00

- **WH:**

- affection:  $1 + \log_{10}(20) \approx 1 + 1.3010 = 2.30$
- jealous:  $1 + \log_{10}(11) \approx 1 + 1.0414 = 2.04$
- gossip:  $1 + \log_{10}(6) \approx 1 + 0.7782 = 1.78$
- wuthering:  $1 + \log_{10}(38) \approx 1 + 1.5798 = 2.58$

### 5.4.3 Step 3: Euclidean Norms and $L_2$ Unit Normalization

Computing  $L_2$  vector lengths:

$$\|\vec{v}_{\text{SaS}}\|_2 = \sqrt{3.06^2 + 2.00^2 + 1.30^2 + 0^2} = \sqrt{15.0536} \approx \mathbf{3.88}, \quad (5.14)$$

$$\|\vec{v}_{\text{PaP}}\|_2 = \sqrt{2.76^2 + 1.85^2 + 0^2 + 0^2} = \sqrt{11.0401} \approx \mathbf{3.32}, \quad (5.15)$$

$$\|\vec{v}_{\text{WH}}\|_2 = \sqrt{2.30^2 + 2.04^2 + 1.78^2 + 2.58^2} = \sqrt{19.2764} \approx \mathbf{4.39}. \quad (5.16)$$

Dividing by vector norms produces unit-normalized vectors:

| Term             | SaS ( $\vec{v}/3.88$ )             | PaP ( $\vec{v}/3.32$ )             | WH ( $\vec{v}/4.39$ )              |
|------------------|------------------------------------|------------------------------------|------------------------------------|
| <b>affection</b> | $3.06/3.88 \approx \mathbf{0.789}$ | $2.76/3.32 \approx \mathbf{0.832}$ | $2.30/4.39 \approx \mathbf{0.524}$ |
| <b>jealous</b>   | $2.00/3.88 \approx \mathbf{0.515}$ | $1.85/3.32 \approx \mathbf{0.555}$ | $2.04/4.39 \approx \mathbf{0.465}$ |
| <b>gossip</b>    | $1.30/3.88 \approx \mathbf{0.335}$ | $0.00/3.32 = \mathbf{0.000}$       | $1.78/4.39 \approx \mathbf{0.405}$ |
| <b>wuthering</b> | $0.00/3.88 = \mathbf{0.000}$       | $0.00/3.32 = \mathbf{0.000}$       | $2.58/4.39 \approx \mathbf{0.588}$ |

Table 5.2: Unit-normalized ( $L_2$ ) document vectors across the three novels.

### 5.4.4 Step 4: Cosine Similarity Calculations

$$\cos(\text{SaS}, \text{PaP}) = (0.789 \cdot 0.832) + (0.515 \cdot 0.555) \approx \mathbf{0.94}, \quad (5.17)$$

$$\cos(\text{SaS}, \text{WH}) = (0.789 \cdot 0.524) + (0.515 \cdot 0.465) + (0.335 \cdot 0.405) \approx \mathbf{0.79}, \quad (5.18)$$

$$\cos(\text{PaP}, \text{WH}) = (0.832 \cdot 0.524) + (0.555 \cdot 0.465) \approx \mathbf{0.69}. \quad (5.19)$$

### Result Interpretation

$$\cos(\text{SaS}, \text{PaP}) \approx 0.94 > \cos(\text{SaS}, \text{WH}) \approx 0.79 > \cos(\text{PaP}, \text{WH}) \approx 0.69. \quad (5.20)$$

Cosine Similarity reflects stylistic coherence: Jane Austen’s two works exhibit 94% proximity, markedly higher than the cross-author comparison with Emily Brontë.

## 5.5 Weighting Variations and SMART Notation

Gerard Salton and the Cornell SMART team systematized term-weighting variants into a standard mnemonic triplet notation:

$$\text{SMART Scheme} \implies ddd.qqq, \quad (5.21)$$

where the first triplet *ddd* defines document weighting and the second triplet *qqq* defines query weighting.

| Term Frequency ( <i>tf</i> )             | Document Frequency ( <i>df</i> )                    | Normalization                                |
|------------------------------------------|-----------------------------------------------------|----------------------------------------------|
| <b>n</b> <i>tf</i> (natural)             | <b>n</b> 1 (no idf)                                 | <b>n</b> None (1)                            |
| <b>l</b> $1 + \log_{10}(tf)$             | <b>t</b> $\log_{10}(N/df)$                          | <b>c</b> Cosine ( $1/\ \vec{w}\ _2$ )        |
| <b>a</b> $0.5 + 0.5 \frac{tf}{\max(tf)}$ | <b>p</b> $\max\left(0, \log \frac{N-df}{df}\right)$ | <b>u</b> Pivoted unique                      |
| <b>b</b> 1 if <i>tf</i> > 0, else 0      |                                                     | <b>b</b> Byte size ( $1/\text{len}^\alpha$ ) |

Table 5.3: Systematic taxonomy of weighting components in SMART notation.

### 5.5.1 The Standard Industrial Baseline: **lnc.ltc**

The asymmetric combination **lnc.ltc** is one of the most effective and widely adopted configurations:

- **For Document (lnc):**
  - **l**: Logarithmic Term Frequency ( $1 + \log_{10} tf$ ).
  - **n**: No IDF factor applied to document vectors.
  - **c**: Cosine ( $L_2$ ) length normalization.

- **For Query (1tc):**
  - 1: Logarithmic Term Frequency ( $1 + \log_{10} tf$ ).
  - t: Classical IDF weighting  $\log_{10}(N/df)$ .
  - c: Cosine ( $L_2$ ) length normalization.

#### Architectural Rationale for Asymmetry

Omitting IDF from document weights (**n**) prevents corrupting the index with static global stats and avoids penalizing long documents that legitimately repeat topic terms. The burden of discriminating term importance is assigned entirely to the query vector (**t**).

### 5.5.2 Full 1nc.1tc Numerical Worked Example

Setup:

- **Document:** “*car insurance auto insurance*”
- **Query:** “*best car insurance*”
- $N = 1\,000\,000$ . Document frequencies:
  - auto:  $df = 5\,000 \implies idf = \log_{10}(10^6/5000) \approx 2.3$
  - best:  $df = 50\,000 \implies idf = \log_{10}(10^6/50000) \approx 1.3$
  - car:  $df = 10\,000 \implies idf = \log_{10}(10^6/10000) = 2.0$
  - insurance:  $df = 1\,000 \implies idf = \log_{10}(10^6/1000) = 3.0$

#### Step 1: Query Vector Processing (1tc)

| Term      | $tf$ raw | $tf$ -wt ( $1 + \log tf$ ) | $df$   | $idf$ | Unnorm. weight         | Norm. weight ( $w/\ \vec{q}\ $ ) |
|-----------|----------|----------------------------|--------|-------|------------------------|----------------------------------|
| auto      | 0        | 0.0                        | 5 000  | 2.3   | $0.0 \times 2.3 = 0.0$ | <b>0.00</b>                      |
| best      | 1        | 1.0                        | 50 000 | 1.3   | $1.0 \times 1.3 = 1.3$ | $1.3/3.84 \approx$ <b>0.34</b>   |
| car       | 1        | 1.0                        | 10 000 | 2.0   | $1.0 \times 2.0 = 2.0$ | $2.0/3.84 \approx$ <b>0.52</b>   |
| insurance | 1        | 1.0                        | 1 000  | 3.0   | $1.0 \times 3.0 = 3.0$ | $3.0/3.84 \approx$ <b>0.78</b>   |

Table 5.4: Query vector computation under the 1tc scheme.

Query Euclidean length:

$$\|\vec{q}\|_2 = \sqrt{0^2 + 1.3^2 + 2.0^2 + 3.0^2} = \sqrt{14.69} \approx \mathbf{3.84}. \quad (5.22)$$

## Step 2: Document Vector Processing (lnc)

| Term      | <i>tf</i> raw | <i>tf</i> -wt ( $1 + \log tf$ ) | <i>idf</i> (none for n) | Unnorm. weight | Norm. weight ( $w/\ \vec{d}\ $ ) |
|-----------|---------------|---------------------------------|-------------------------|----------------|----------------------------------|
| auto      | 1             | 1.0                             | —                       | 1.0            | $1.0/1.92 \approx \mathbf{0.52}$ |
| best      | 0             | 0.0                             | —                       | 0.0            | <b>0.00</b>                      |
| car       | 1             | 1.0                             | —                       | 1.0            | $1.0/1.92 \approx \mathbf{0.52}$ |
| insurance | 2             | $1 + \log_{10}(2) \approx 1.30$ | —                       | 1.3            | $1.3/1.92 \approx \mathbf{0.68}$ |

Table 5.5: Document vector computation under the lnc scheme.

Document Euclidean length:

$$\|\vec{d}\|_2 = \sqrt{1.0^2 + 0^2 + 1.0^2 + 1.30^2} = \sqrt{3.69} \approx \mathbf{1.92}. \quad (5.23)$$

## Step 3: Final Scored Ranking

Computing the dot product between unit vectors:

$$\begin{aligned}
 \text{Score}(q, d) &= \vec{q}_{\text{norm}} \cdot \vec{d}_{\text{norm}} \\
 &= (0.00 \cdot 0.52) + (0.34 \cdot 0.00) + (0.52 \cdot 0.52) + (0.78 \cdot 0.68) \\
 &= 0 + 0 + 0.2704 + 0.5304 \approx \mathbf{0.80}.
 \end{aligned} \quad (5.24)$$

## 5.6 Key Review Questions and Core Concepts

1. **Why does Jaccard similarity sometimes score less relevant documents higher?** It operates on raw sets, ignoring within-document term frequency (*tf*) and collection-wide rarity (*idf*). It also penalizes longer documents via denominator union growth.
2. **Why is Euclidean distance unsuitable for ranking in the Vector Space Model?** Euclidean distance is sensitive to vector magnitude: duplicate documents with identical topics have twice the length, placing them far from short queries despite zero angular deviation ( $\theta = 0^\circ$ ).
3. **What is the geometric interpretation of  $L_2$  normalization?** It radially projects vectors onto the unit hypersphere, neutralizing document length variations and simplifying cosine similarity to a dot product.
4. **What is the effect of IDF on a single-term query?** Zero effect on relative document ranking: IDF acts as a constant scaling factor for all matching documents, leaving relative order unchanged.

## 5.7 Best Match 25 (Okapi BM25)

Developed by Stephen Robertson and Karen Spärck Jones at City University London, **BM25** remains the premier state-of-the-art non-linear lexical baseline:

$$\text{Score}_{\text{BM25}}(D, Q) = \sum_{i=1}^n \text{IDF}(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{\text{avgdl}}\right)}, \quad (5.25)$$

with Robertson-Spärck Jones IDF:

$$\text{IDF}(q_i) = \ln \left( \frac{N - n(q_i) + 0.5}{n(q_i) + 0.5} + 1 \right). \quad (5.26)$$

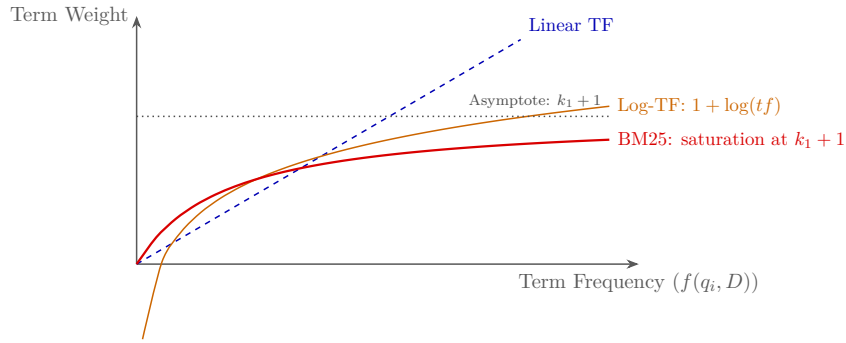


Figure 5.2: Qualitative comparison of linear TF, log-TF, and BM25 asymptotic saturation.

- $k_1 \in [1.2, 2.0]$  governs term frequency saturation, bounded by  $\lim_{f \rightarrow \infty} = k_1 + 1$ .
- $b \in [0, 1]$  (typically 0.75) governs relative document length normalization against average length avgdl.

### Large-Scale BM25 Evaluation

Okapi BM25 defines the scoring function for individual query–document pairs. On web-scale collections containing billions of documents, evaluating BM25 across inverted lists using modern query processing strategies (*Term-at-a-Time* and *Document-at-a-Time* with WAND dynamic pruning) is detailed in Chapter 6.

## 5.8 Summary Conceptual Map

Figure 5.3 synthesizes the full trajectory from raw text acquisition to vector and probabilistic scoring.

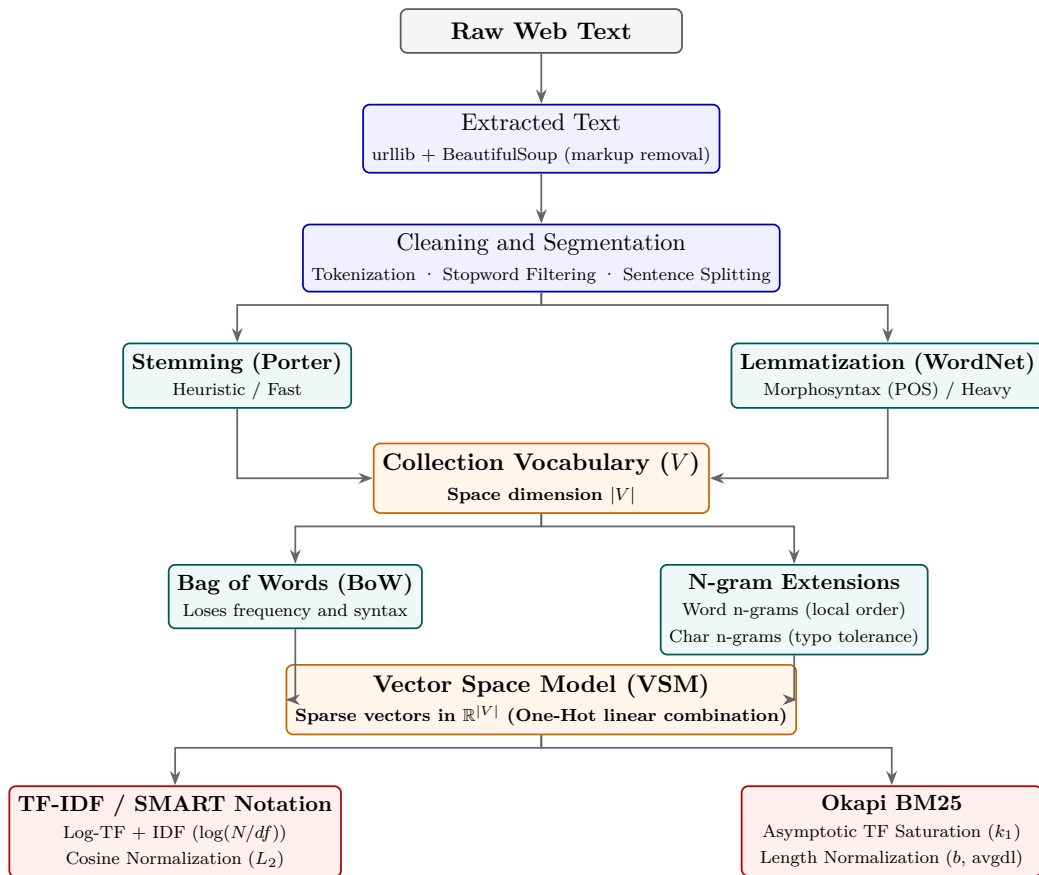


Figure 5.3: Integrated conceptual map of the IR pipeline: from text scraping to vector and probabilistic retrieval models.

### From Mathematical Scoring to Search Engine Architecture

With the mathematical formalization of the Vector Space Model, SMART schemes, and Okapi BM25, the principles of *what* must be computed are established. Chapter 6 investigates the architectural and systems dimension of *how* to evaluate these continuous scores within milliseconds across web-scale inverted indexes.

# Chapter 6

## Search Engine Architecture: Indexing and Query Processing

The preceding chapters established the core mathematical foundations of Information Retrieval: evaluation metrics for ranked lists (Chapter 1), linguistic properties of textual data (Chapter 3), and the mathematical derivation of relevance scoring under the Vector Space Model and Okapi BM25 (Chapter 5). These models define *what* must be computed to rank documents with respect to user queries.

This chapter addresses the complementary systems and algorithmic challenge: **how to compute these continuous scores in real time over planetary-scale corpora**, maintaining sub-50 ms latencies across collections containing billions of documents. We examine the two-stage retrieval pipeline, analyze the extreme sparsity of term–document matrices contrasting *Forward Indexes* with *Inverted Indexes*, study memory layout optimizations and integer compression, inspect positional indexes for phrase queries alongside query expansion pathologies, and deliver a detailed algorithmic and architectural comparison between *Term-at-a-Time* (TAAT) and *Document-at-a-Time* (DAAT) query processing.

### 6.1 System Architecture and the Two-Stage Retrieval Paradigm

Large-scale search engines must satisfy stringent throughput and latency constraints ( $< 50\text{--}100$  ms). To achieve this, computation is strictly divided between massive offline batch processes and online interactive query servicing.

### 6.1.1 The Offline Pipeline: Ingestion, Indexing, and Compression

The offline pipeline processes an unstructured corpus  $\mathcal{D} = \{d_1, d_2, \dots, d_N\}$  through three stages:

1. **Text Processing and Parsing:** Web crawled documents are stripped of markup, tokenized, and normalized (Section 3.3.2).
2. **Vocabulary and Index Construction:** The lexicon  $\mathcal{V} = \{t_1, \dots, t_M\}$  is extracted, mapping every unique term to its occurrences across documents.
3. **Lossless Integer Compression:** Because raw index data can rival or exceed the raw text size, integer compression algorithms are applied to ensure index segments reside in high-speed memory tiers (RAM and CPU caches, Chapter 2).

### 6.1.2 The Two-Stage Retrieval Pipeline

Commercial web search engines resolve queries using a two-stage pipeline:

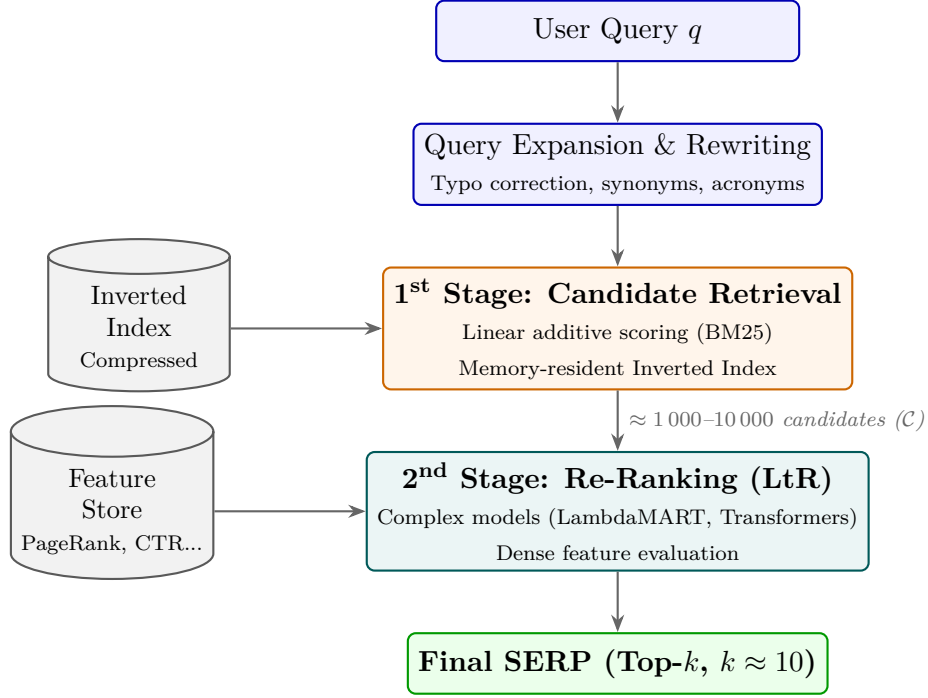


Figure 6.1: Architecture of the two-stage retrieval pipeline: high-throughput coarse candidate generation followed by high-capacity machine-learned re-ranking.

1. **First Stage (Candidate Retrieval):** Operates over the entire corpus ( $N \sim 10^9\text{--}10^{10}$  documents) using fast additive scoring functions (Okapi BM25) to isolate a small candidate set:

$$\mathcal{C} \subset \mathcal{D}, \quad |\mathcal{C}| \approx 1\,000\text{--}10\,000 \text{ documents.} \quad (6.1)$$

2. **Second Stage (Re-Ranking / Heavy Scoring):** Operates exclusively on  $\mathcal{C}$  using complex Machine Learning models (Learning-to-Rank algorithms such as LambdaMART [4] or cross-encoder Transformers) evaluating hundreds to thousands of dense signals to produce the final top- $k$  ( $k \approx 10$ ).

## 6.2 Scoring Models: The Dot Product Bridge to Dense Retrieval

### 6.2.1 Recap: Limitations of Exact Boolean and Jaccard Scoring

As established in Chapter 4 (Sections 4.2 and 4.3), exact Boolean matching suffers from the *Feast or Famine* pathology, while unweighted set-based Jaccard scoring unfairly penalizes longer documents and ignores term frequency (*tf*) and collection specificity (*idf*).

### 6.2.2 The Dot Product as a Unified Scoring Operator

The Vector Space Model and statistical weighting schemes resolve these limitations by formulating relevance as a vector dot product:

$$\text{Score}(q, d) = \mathbf{q} \cdot \mathbf{d} = \sum_{t \in \mathcal{V}} w(t, q) \cdot w(t, d) = \sum_{t \in q \cap d} w(t, q) \cdot w(t, d). \quad (6.2)$$

This formulation bridges classical lexical retrieval and modern neural dense retrieval:

- **Lexical Sparse Retrieval:** Vectors reside in  $\mathbb{R}^{|\mathcal{V}|}$ , with dimensions corresponding to exact vocabulary terms ( $|\mathcal{V}| \sim 10^7$ ). Vectors are ultra-sparse, scored using statistical weights (TF-IDF, BM25).
- **Dense Neural Retrieval:** Transformer-based encoders project queries and passages into low-dimensional dense embeddings  $\mathbf{v}_q, \mathbf{v}_d \in \mathbb{R}^k$  ( $k \approx 768$ – $1536$ ). Semantic similarity is determined by vector dot product:

$$\text{Score}_{\text{neural}}(q, d) = \mathbf{v}_q \cdot \mathbf{v}_d = \sum_{i=1}^k v_{q,i} \cdot v_{d,i}. \quad (6.3)$$

In both domains, real-time efficiency depends on aggregating dot products over massive corpora.

## 6.3 High-Performance Data Structures: Forward Index vs. Inverted Index

### 6.3.1 Matrix Sparsity and the Dense Bottleneck

For a web-scale search engine:

- Vocabulary size:  $|\mathcal{V}| \sim 10^7$  unique terms.
- Document collection size:  $|\mathcal{D}| \sim 10^9\text{--}10^{10}$  documents.
- Total matrix cells:  $|\mathcal{V}| \times |\mathcal{D}| \approx 10^{16}\text{--}10^{17}$  cells.

Because an average document contains only  $10^2\text{--}10^3$  distinct words, the non-zero density is below 0.001%; over 99.99% of matrix entries are identically zero. Allocating a dense matrix would require petabytes of redundant memory and demand an unacceptable  $\mathcal{O}(|\mathcal{V}| \cdot |\mathcal{D}|)$  scan per query.

### 6.3.2 Sparse Serialization: Forward Index vs. Inverted Index

Sparse representations record only non-zero (ID, Value) pairs:

- **Forward Index (Column-Wise / Document-Oriented):** Maps each document to its constituent terms:  $\text{DocID} \rightarrow [(t_a, w_a), \dots]$ . While memory-efficient, evaluating a query still requires inspecting all  $N$  documents ( $\mathcal{O}(|\mathcal{D}|)$ ).
- **Inverted Index (Row-Wise / Term-Oriented):** Maps each term to the ordered sequence of documents containing it:  $t \rightarrow [(\text{DocID}_i, w_i), \dots]$ . This enables instantaneous static pruning: documents containing none of the query terms are skipped entirely without executing CPU cycles.

| Property                | Forward Index                                        | Inverted Index                                                                       |
|-------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Orientation</b>      | Column-wise<br>(Document-Oriented).                  | Row-wise<br>(Term-Oriented).                                                         |
| <b>Mapping</b>          | $\text{DocID} \rightarrow [(t_a, w_a), \dots]$       | $t \rightarrow [(\text{DocID}_i, w_i), \dots]$                                       |
| <b>Query Complexity</b> | $\mathcal{O}( \mathcal{D} )$ (full collection scan). | $\mathcal{O}\left(\sum_{t \in q}  \mathcal{L}(t) \right)$ (candidate postings only). |
| <b>Engine Function</b>  | Document metadata storage, snippet generation.       | Primary indexing core for candidate retrieval.                                       |

Table 6.1: Systematic comparison between Forward Index and Inverted Index.

## 6.4 Inverted Index Anatomy, Memory Layout, and Compression

### 6.4.1 The Lexicon (Dictionary)

The lexicon stores the vocabulary  $\mathcal{V}$ , recording for each term  $t$  its text string, document frequency  $df_t$  (for precomputed IDF), and memory pointer or disk offset to its posting list. The lexicon is kept memory-resident in **RAM** using fast structures (B-Trees, Trie, or front-coded arrays).

### 6.4.2 Posting Lists: Physical Stream Decoupling

Conceptually, a posting list is a sequence of tuples:

$$\mathcal{L}(t) = [\langle d_1, \text{tf}_1 \rangle, \langle d_2, \text{tf}_2 \rangle, \dots]. \quad (6.4)$$

In high-performance implementations, storing interleaved structs wastes compression potential. Engines decouple posting lists into two distinct physical streams:

1. **DocID Stream (Strictly Monotonic):**  $\mathbf{D}_t = [\text{DocID}_1, \dots, \text{DocID}_m]$  with  $\text{DocID}_i < \text{DocID}_{i+1}$ . Storing successive differences ( $d$ -gaps):

$$\Delta_1 = \text{DocID}_1, \quad \Delta_i = \text{DocID}_i - \text{DocID}_{i-1} \quad (i \geq 2) \quad (6.5)$$

yields small integers compressed with methods such as Variable Byte, Elias-Fano, PForDelta, or interpolative coding [3].

2. **Term Frequency Stream (Unordered):**  $\mathbf{F}_t = [\text{tf}_1, \dots, \text{tf}_m]$ . Values are small positive integers (1, 2, 3) compressed using symmetric variable-length integer codes.

### 6.4.3 Compressed Integers vs. Precomputed Floats

Search engines universally store **compressed integers** rather than precomputed floating-point scores. Large-scale search systems are **memory-bound** (Chapter 2): RAM-to-CPU transfer latencies ( $\approx 80$  ns) dominate execution time. Compressing postings drastically reduces memory bandwidth demands; the minor CPU arithmetic required to compute BM25 scores on the fly is negligible compared to memory wait cycles.

## 6.5 Query Types, Positional Indexes, and Query Expansion

### 6.5.1 Phrase Queries and Positional Indexes

Phrase queries (e.g., "neural network architecture") require terms to appear adjacent ( $\text{pos}(t_{i+1}) = \text{pos}(t_i) + 1$ ). Frequency-only inverted indexes cannot verify this constraint. The engine extends the posting structure into a **Positional Index**:

$$t \longrightarrow \left[ \langle \text{DocID}_k, \text{tf}_k, [\text{pos}_1, \dots, \text{pos}_{\text{tf}}] \rangle \right]. \quad (6.6)$$

Positional tracking incurs a  $2 \times -3 \times$  storage overhead over non-positional indexes.

### 6.5.2 Query Expansion and Semantic Drift

Engines expand raw user queries with synonyms, spell corrections, and acronyms. Unconstrained lexical expansion can cause severe *semantic drift*. A classic example in Italian occurs when expanding "babbo francesco":

1. *Babbo* is expanded to its regional synonym *papà* (father).
2. Normalization strips accents, mapping *papà* to *papa* (Pope).
3. The query becomes:

("babbo" OR "pap`a" OR "papa") AND "francesco"

4. High PageRank pages for Pope Francis (*Papa Francesco*) overwhelm relevance, making Wikipedia's biography of the Pontiff the top result.

### 6.5.3 Stemming vs. Query Expansion

While stemming and lemmatization reduce words along morphological lines (e.g., *connecting*  $\rightarrow$  *connect*, Section 3.3.3), they cannot link etymologically distinct synonyms (such as *babbo* and *papà*). Modern engines avoid aggressive indexing-time stemming and instead resolve synonyms dynamically via contextual neural query expansion.

## 6.6 Query Processing: Term-at-a-Time (TAAT) vs. Document-at-a-Time (DAAT)

### 6.6.1 Term-at-a-Time (TAAT)

TAAT iterates through posting lists **sequentially, one term at a time**:

1. Allocates a global accumulator array  $A[\text{DocID}] \in \mathbb{R}$ .
2. Reads the first posting list  $\mathcal{L}(t_1)$ , updating accumulator scores.
3. Proceeds to  $\mathcal{L}(t_2)$ , accumulating scores until all query terms are exhausted.
4. Extracts the top- $k$  entries using a min-heap over  $A$ .

---

#### Algorithm 1: Term-at-a-Time (TAAT)

---

**Input:** Query  $q = \{t_1, \dots, t_m\}$ , Inverted Index, integer  $k$   
**Output:** Min-Heap  $H$  of Top- $k$  scored documents

```

1  $A \leftarrow \text{InitializeAccumulatorTable}()$ 
2 foreach term  $t_i \in q$  do
3    $\mathcal{L}_i \leftarrow \text{FetchPostingList}(t_i)$ 
4    $\text{idf}_i \leftarrow \text{ComputeIDF}(t_i)$ 
5   foreach  $\langle d, \text{tf} \rangle \in \mathcal{L}_i$  do
6      $s_{\text{partial}} \leftarrow \text{EvaluateTermScore}(\text{tf}, \text{idf}_i)$ 
7     if  $d \in A$  then
8        $A[d] \leftarrow A[d] + s_{\text{partial}}$ 
9     else
10       $A[d] \leftarrow s_{\text{partial}}$ 
11  $H \leftarrow \text{InitializeMinHeap}(\text{capacity} = k)$ 
12 foreach entry  $(d, \text{score}) \in A$  do
13   if  $H.\text{Size}() < k$  then
14      $H.\text{Insert}(d, \text{score})$ 
15   else
16     if  $\text{score} > H.\text{MinScore}()$  then
17        $H.\text{ExtractMin}()$ 
18        $H.\text{Insert}(d, \text{score})$ 
19 return  $H$ 

```

---

**Characteristics.** High sequential memory locality and hardware prefetching, but incurs a heavy RAM accumulator footprint and cannot perform dynamic early pruning.

### 6.6.2 Document-at-a-Time (DAAT)

DAAT processes all query posting lists **in parallel across synchronized pointers**:

1. Initializes cursors  $p_i$  at the head of each list  $\mathcal{L}(t_i)$ .
2. Identifies the current minimum DocID across cursors:

$$d_{\min} = \min_i \{p_i.\text{CurrentDocID}()\}. \quad (6.7)$$

3. Accumulates the full final score  $S(q, d_{\min})$  across all cursors matching  $d_{\min}$ .
4. Updates the fixed-size top- $k$  min-heap immediately.
5. Advances all cursors matching  $d_{\min}$  to their next entry.

---

**Algorithm 2:** Document-at-a-Time (DAAT)

---

**Input:** Query  $q = \{t_1, \dots, t_m\}$ , Inverted Index, integer  $k$

**Output:** Min-Heap  $H$  of Top- $k$  scored documents

```

1 cursors  $\leftarrow$  InitializeListCursors( $q$ )
2  $H \leftarrow$  InitializeMinHeap(capacity =  $k$ )
3 while HasActiveCursors( $cursors$ ) do
4    $d_{\min} \leftarrow \min\{p.\text{CurrentDocID}() \mid p \in \text{cursors} \wedge p.\text{IsActive}()\}$ 
5    $score_{\text{tot}} \leftarrow 0.0$ 
6   foreach  $p \in \text{cursors}$  do
7     if  $p.\text{IsActive}() \wedge p.\text{CurrentDocID}() == d_{\min}$  then
8        $score_{\text{tot}} \leftarrow$ 
9          $score_{\text{tot}} + \text{EvaluateTermScore}(p.\text{CurrentTF}(), p.\text{Term}())$ 
10         $p.\text{Advance}()$ 
11   if  $H.\text{Size}() < k$  then
12      $H.\text{Insert}(d_{\min}, score_{\text{tot}})$ 
13   else
14     if  $score_{\text{tot}} > H.\text{MinScore}()$  then
15        $H.\text{ExtractMin}()$ 
16        $H.\text{Insert}(d_{\min}, score_{\text{tot}})$ 
17 return  $H$ 

```

---

*Characteristics:* Minimal working memory ( $\mathcal{O}(k)$  heap and  $\mathcal{O}(m)$  cursor state) and native support for dynamic early termination algorithms (WAND and Block-Max WAND), at the cost of non-contiguous interleaved memory access patterns.

| Dimension                  | Term-at-a-Time (TAAT)                                           | Document-at-a-Time (DAAT)                          |
|----------------------------|-----------------------------------------------------------------|----------------------------------------------------|
| <b>Traversal Direction</b> | Horizontal (term-by-term, sequential).                          | Vertical (document-by-document, synchronous).      |
| <b>Score State</b>         | Partial progressive accumulators.                               | Fully consolidated score per document.             |
| <b>RAM Footprint</b>       | High ( $\mathcal{O}( \bigcup \mathcal{L}(t_i) )$ accumulators). | Minimal ( $\mathcal{O}(k)$ for Top- $k$ min-heap). |
| <b>Cache Locality</b>      | Excellent (contiguous stream scans).                            | Interleaved (pointer jumping across lists).        |
| <b>Early Termination</b>   | Ineffective.                                                    | Native and powerful (WAND, Block-Max).             |
| <b>Industry Role</b>       | GPU and SIMD accelerator architectures.                         | Core CPU search engines (Lucene, Elasticsearch).   |

Table 6.2: Systematic comparison between TAAT and DAAT query processing strategies.

### 6.6.3 Significance in Modern Vector and Hybrid Search

The TAAT vs. DAAT paradigm extends directly into modern retrieval systems:

- **Sparse Neural Retrieval (e.g., SPLADE [6]):** Employs inverted indexes over expanded neural vocabularies evaluated using DAAT with Block-Max WAND pruning.
- **Hardware Acceleration:** GPU-based vector accelerators leverage TAAT-style batching to exploit streaming multiprocessors and high memory bandwidth.
- **Dedicated Vector Engines (e.g., Qdrant):** Modern engines combine approximate nearest neighbor graphs ( $k$ -ANN on HNSW) with inverted indexes for payload filtering, combining DAAT and TAAT principles for hybrid multi-stage retrieval.

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